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TECHNICAL SPECIFICATION

Core Network and Interoperability Testing (INT); Conformance Testing for IP/ICMP Translation Algorithm; (IETF RFC 7915); Part 2: Test Suite Structure (TSS) and Test Purposes (TP)

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Contents

Intellectual Property Rights 4
Foreword 4
Modal verbs terminology 4
1 Scope 5
2 References 5
2.1 Normative references 5
2.2 Informative references 5
3 Definition of terms, symbols and abbreviations 6
3.1 Terms 6
3.2 Symbols 6
3.3 Abbreviations 6
4 Test configurations 7
4.1 Introduction 7
4.2 Test configuration 7
5 Test Suite Structure (TSS) and Test Purposes (TP) 7
5.1 Test Suite Structure 7
5.1.1 TP naming convention 7
5.1.2 Test strategy 8
5.1.3 TP structure 8
5.2 Test Purposes 9
5.2.1 PICS references 9
5.2.2 Translating from IPv4 to IPv6 9
5.2.2.1 Translating IPv4 Headers to IPv6 Headers 9
5.2.2.2 Translating ICMPv4 Headers to ICMPv6 Headers 15
5.2.2.3 Translating ICMPv4 Error Messages into ICMPv6 70
5.2.2.4 Generation of ICMPv4 Error Message 73
5.2.2.5 Transport-Layer Header Translation 74
5.2.2.6 Knowing When to Translate 77
5.2.3 Translating from IPv6 to IPv4 77
5.2.3.1 Translating IPv6 Headers to IPv4 Headers 77
5.2.3.2 Translating ICMPv6 Headers into ICMPv4 Headers 85
5.2.3.3 Translating ICMPv6 Error Messages into ICMPv4 110
5.2.3.4 Generation of ICMPv6 Error Messages 113
5.2.3.5 Transport-Layer Header Translation 113
5.2.3.6 Knowing When to Translate 115
History 116

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Foreword

This Technical Specification (TS) has been produced by ETSI Technical Committee Core Network and Interoperability Testing (INT).

The present document is part 2 of a multi-part deliverable. Full details of the entire series can be found in part 1 [2].

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

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1 Scope

The present document provides the Test Suite Structure (TSS) and Test Purposes (TP) for the test specification for the IP/ICMP Translation Algorithm as specified in IETF RFC 7915 [1] in compliance with the relevant requirements and in accordance with the relevant guidance given in ISO/IEC 9646-7 [i.2] and ETSI ETS 300 406 [i.3].

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

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The following referenced documents are necessary for the application of the present document.

[1] IETF RFC 7915: "IP/ICMP Translation Algorithm".

[2] ETSI TS 104 153-1: "Core Network and Interoperability Testing (INT); Conformance Testing for IP/ICMP Translation Algorithm; (IETF RFC 7915); Part 1: Protocol Implementation Conformance Statement (PICS)".

[3] IETF RFC 1191: "Path MTU Discovery".

[4] IETF RFC 4884: "Extended ICMP to Support Multi-Part Messages".

[5] IETF RFC 950: "Internet Standard Subnetting Procedure".

[6] IETF RFC 1256: "ICMP Router Discovery Messages".

2.2 Informative references

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The following referenced documents may be useful in implementing an ETSI deliverable or add to the reader's understanding, but are not required for conformance to the present document.

[i.1] ISO/IEC 9646-1: "Information technology — Open Systems Interconnection — Conformance testing methodology and framework — Part 1: General concepts".

[i.2] ISO/IEC 9646-7: "Information technology — Open Systems Interconnection — Conformance testing methodology and framework — Part 7: Implementation Conformance Statements".

[i.3] ETSI ETS 300 406: "Methods for testing and Specification (MTS); Protocol and profile conformance testing specifications; Standardization methodology".

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in IETF RFC 7915 [1] and the following apply:

Abstract Test Method (ATM): Refer to ISO/IEC 9646-1 [i.1].

Abstract Test Suite (ATS): Refer to ISO/IEC 9646-1 [i.1].

Implementation Under Test (IUT): Refer to ISO/IEC 9646-1 [i.1].

Test Purpose (TP): Refer to ISO/IEC 9646-1 [i.1].

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in IETF RFC 7915 [1] and the following apply:

3GPP	3rd Generation Partnership Project
AH	Authentication Header
ATS	Abstract Test Suite
DF	Don't Fragment
ESP	Encapsulating Security Payload
GE	Generation of ICMPv4/ICMPv6 Error message
ICMP	Internet Control Message Protocol
ICMPE	translating ICMPv4/ICMPv6 Error messages
ICMPH	translating ICMPv4/ICMPv6 Headers
ICMPv4	Internet Control Message Protocol version 4
ICMPv6	Internet Control Message Protocol version 6
IP	Internet Protocol
IPHDR	translating IPv4/IPv6 HeaDers
IPv4	Internet Protocol version 4
IPv6	Internet Protocol version 6
IUT	Implement Under Test
MF	More Fragment
MLD	Multicast Listener Discovery
MTU	Maximum Transport Unit
PICS	Protocol Implementation Conformance Statement
SIIT	Stateless IP/ICMP Translation
TCP	Transmission Control Protocol
TLH	Transport-Layer Header translation
TOS	Type Of Service
TP	Test Purpose
TS	Test System
TSS	Test Suite Structure
TTL	Time To Live
UDP	User Datagram Protocol
WTT	knowing When To Translate
XLAT	Translator

4 Test configurations

4.1 Introduction

Test purposes of the present document address the IP/ICMP translators that is the implementation of stateless IP/ICMP translating algorithm.

4.2 Test configuration

Following configurations are simplified to highlight tested interface and involved entities.

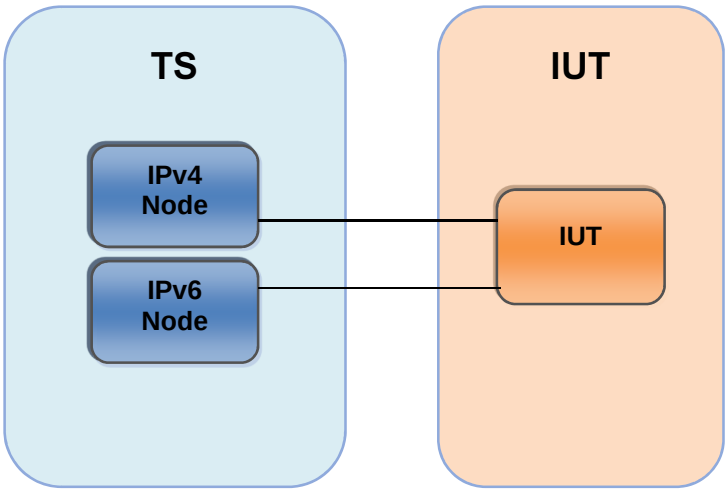


Figure 1: Test configuration CF_XLAT_SIIT

5 Test Suite Structure (TSS) and Test Purposes (TP)

5.1 Test Suite Structure

5.1.1 TP naming convention

TPs are numbered, starting at 01, within each group. Groups are organized according to the TSS.

Table 1: TP identifier naming convention scheme

Identifier: **<TP>_<scope>_<nn>**		
<tp> = Test Purpose:		fixed to "TP"
<interface or protocol>		Interface or protocol: stateless IP/ICMP translation (SIIT) algorithm
<direction>		from IPv4 to IPv6 (T46) or from IPv6 to IPv4 (T64)
<scope> = group		IPHDR Translating IPv4/IPv6 Headers
		ICMPH Translating ICMPv4/ICMPv6 Headers
		ICMPE Translating ICMPv4/ICMPv6 Error Messages
		GE Generation of ICMPv4/ICMPv6 Error Message
		TLH Transport-Layer Header Translation
		WTT Knowing When to Translate
<nn> = sequential number		(01 to 99)

5.1.2 Test strategy

As the base specification in IETF RFC 7915 [1] contains no explicit requirements for testing, the TPs were generated as a result of an analysis of the base standard and the PICS specification ETSI TS 104 153-1 [2].

5.1.3 TP structure

Each TP has been written in a manner which is consistent with all other TPs. The intention of this is to make the TPs more readable and checkable. A particular structure has been used which is illustrated in Table 2. Table 2 should be read in conjunction with any TP, i.e. use a TP as an example to facilitate the full comprehension of Table 2.

Table 2: Structure of a single TP

TP part	Text	Example
Header	**<Identifier>** **<clause number in base IETF RFC 7915 [**1**]>** **<PICS reference>**	**see Table 1** **clause 8.3.11.2** **A.3/3.1**
Summary	*Short free text description of the test objective*	Verify that the IUT can send a HANDOVER REQUIRED message containing mandatory IEs due to UE mobility management procedure.
Configuration	*Test configuration as described in clause 4.2*	CF_XLAT_SIIT
Initial condition (optional)	*Free text description of the condition that the IUT has reached before the test purpose applies.*	
Start point	Ensure that the IUT in the <state> *see IETF RFC 7915 [1], clause 8.1* *and/or further actions before stimulus* *if the action is sending/receiving* *see below for message structure*	Handover Preparation having sent a HANDOVER_REQUIRED
Stimulus	<**trigger**>, *see below for message structure* *or* <goal>	on receipt of a IPv4 packet (see note 2)
Reaction	<**action**>. *if the action is sending* *see below for message structure* <next action>, *etc.*	sends, saves, does, etc.
Message structure	<message type> *a)* containing a(n) <packet name> packet (see note 4) *b)* indicating <coding of the field> and *back to a) or b)* (see note 3)	Message exchange, etc. (see note 2)

NOTE 1: Text in italics will not appear in TPs and text between <> is filled in for each TP and may differ from one TP to the next.

NOTE 2: All messages are considered as "valid and compatible" unless otherwise specified in the test purpose. This includes the presence of all mandatory packets as specified in IETF RFC 7915 [1].

NOTE 3: A packet can be embedded into another packet. This is expressed by indentations, e.g. if Message1 contains packet1 and packet2 where packet1 has packet3 embedded this will be expressed like this:

```
sends/receives Message 1
    containing packet1
        containing packet3
            indicating ...
    containing packet2
        indicating ...
```

NOTE 4: Packet value fields used for e.g. identification or address should be equal in the scope of TP if not stated otherwise.

5.2 Test Purposes

5.2.1 PICS references

All PICS items referred to in this clause are as specified in ETSI TS 104 153-1 [2] unless indicated otherwise by another numbered reference. PICS items are only meant for test selection, therefore only PICS items with status optional or conditional are explicitly mentioned.

5.2.2 Translating from IPv4 to IPv6

5.2.2.1 Translating IPv4 Headers to IPv6 Headers

TP Id	TP_SIIT_T46_IPHDR_01
Test Objective	Verify that the IUT successfully adjust the threshold of the minimum IPv6 MTU to a value greater than 1 280 bytes.
Reference	IETF RFC 7915 [1], clause 4.1
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_1_1
Initial Conditions	

with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1350 bytes
}	

Expected Behaviour	

ensure that {
when {
the IUT sets the value of the IPv6 MTU to 1300 bytes
and
receives an IPv4 packet with the length of 1280 bytes containing
DF bit, with the value of 0
from the IPv4 node
}
then {
the IUT sends an IPv6 packet without fragmentation
to the IPv6 node
}
}	

TP Id	TP_SIIT_T46_IPHDR_02
Test Objective	Verify that the IUT successfully sending ICMPv4 "Fragmentation Needed" error message to the IPv4 source address If the DF bit is set and the MTU of the next-hop interface is less than the total length value of the IPv4 packet plus 20.
Reference	IETF RFC 7915 [1], clause 4.1
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_1_2
Initial Conditions	

with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
}	

Expected Behaviour	

ensure that {
when {
IUT receives an IPv4 packet with the length of 1280 bytes containing:
DF bit is set
from the IPv4 node
}
then {
the IUT sends an ICMPv4 "Fragmentation Needed" error message
to the IPv4 node
}
}	

TP Id	TP_SIIT_T46_IPHDR_03
Test Objective	Verify that the IUT successfully translates the fields in the IPv4 header.
Reference	IETF RFC 7915 [1], clause 4.1
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_1_3
Initial Conditions	

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             No Options
             UDP Header
             Data
        from the IPv4 node
    }
    then {
        the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version, with the value of 6
                Traffic Class, with same value of TOS in IPv4 Header
                Flow Label, with the value of all 0
                Payload Length, with the value of total length value from IPv4 header minus the
                                size of the IPv4 header and IPv4 options
                Next Header, with the same value of Protocol in IPv4 Header
                Hop Limit, with the value of Time to Live minus 1
                Source Address, using IPv4 source address and IPv6 prefix to synthesize
                Destination Address, using IPv4 destination address and IPv6 prefix to synthesize
             No Extension Header,
             UDP Header,
             Data
        to the IPv6 node
       }
   }
```

TP Id	TP_SIIT_T46_IPHDR_04
Test Objective	Verify that the IUT successfully translates the fields in the IPv4 header when it ignores the IPv4 TOS and set the traffic class to zero.
Reference	IETF RFC 7915 [1], clause 4.1
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_1_3_2
Initial Conditions	

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS), with value not equal to 0,
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
```

```
                Time to Live (TTL), with the value greater than or equal to 2
                Protocol,
                Header Checksum,
                Source Address,
                Destination Address,
           No Options
           UDP Header
           Data
       from the IPv4 node
    }
    then {
       the IUT sends an IPv6 packet containing
           IPv6 basic header containing
              Traffic Class, with value of all 0
           No Extension Header,
           UDP Header,
           Data
       to the IPv6 node
      }
   }
```

TP Id	TP_SIIT_T46_IPHDR_05
Test Objective	Verify that the IUT successfully translates the fields in the IPv4 header when the Protocol is ICMPv4 (1).
Reference	IETF RFC 7915 [1], clause 4.1
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_1_5

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
       }
```

Expected Behaviour

```
ensure that {
    when {
       IUT receives an IPv4 packet with the total length less than 1280 bytes containing
            IPv4 basic header containing
                Version,
                Internet Header Length (IHL),
                Type of Service (TOS),
                Total Length,
                Identification (Fragment ID),
                Flags,
                Fragment Offset,
                Time to Live (TTL), with the value greater than or equal to 2
                Protocol, with the value of 1
                Header Checksum,
                Source Address,
                Destination Address,
           No Options
           UDP Header
           Data
       from the IPv4 node
    }
    then {
       the IUT sends an IPv6 packet containing
           IPv6 basic header containing
              Next Header, with the value of 58 (ICMPv6)
           No Extension Header,
           UDP Header,
           Data
       to the IPv6 node
      }
   }
```

TP Id	TP_SIIT_T46_IPHDR_06
Test Objective	Verify that the IUT successfully checks for zero sending the ICMPv4 "TTL Exceeded" or ICMPv6 "Hop Limit Exceeded" error after decrementing the TTL or Hop Limit.
Reference	IETF RFC 7915 [1], clause 4.1
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_1_6

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value equal to 1
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             No Options
             UDP Header
             Data
        from the IPv4 node
    }
    then {
        the IUT sends an ICMPv4 "TTL Exceeded" error message
        to the IPv4 node
       }
   }
```

TP Id	TP_SIIT_T46_IPHDR_07
Test Objective	Verify that the IUT silently discards the packet with an illegal source address (e.g. 0.0.0.0, 127.0.0.1, etc.).
Reference	IETF RFC 7915 [1], clause 4.1
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_1_7_1

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol, with the value of 1
                  Header Checksum,
                  Source Address, with the value not equal to 0.0.0.0
                  Destination Address,
             No Options
             UDP Header
             Data
        from the IPv4 node
    }
    then {
        the IUT drops the IPv4 packet
```

```
        }
    }
```

TP Id	TP_SIIT_T46_IPHDR_08
Test Objective	Verify that the IUT successfully translates ICMPv4 Error Messages with an illegal source address (e.g. 0.0.0.0, 127.0.0.1, etc.) into ICMPv6.
Reference	IETF RFC 7915 [1], clause 4.1
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_1_7_1

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol, with the value of 1
                  Header Checksum,
                  Source Address, with the value of 0.0.0.0
                  Destination Address,
             No Options
             UDP Header
             Data
        from the IPv4 node
    }
    then {
        the IUT translates ICMPv4 Error Messages into ICMPv6
        to the IPv6 node
       }
   }
```

TP Id	TP_SIIT_T46_IPHDR_09
Test Objective	Verify that the IUT successfully ignores any options presented in the IPv4 packet.
Reference	IETF RFC 7915 [1], clause 4.1
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_1_4

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             Options containing
                  anyone of options except Source Route
             UDP Header
```

```
            Data
        from the IPv4 node
    }
    then {
        the IUT sends an IPv6 packet without translating IPv4 Options
        to the IPv6 node
       }
   }
```

TP Id	TP_SIIT_T46_IPHDR_10
Test Objective	Verify that the IUT successfully sends an ICMPv4 "Destination Unreachable, Source Route Failed" (Type 3, Code 5) error message when receiving unexpired source route option.
Reference	IETF RFC 7915 [1], clause 4.1
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_1_4_1

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             Options containing
                  Source Route options
             UDP Header
             Data
        from the IPv4 node
    }
    then {
        the IUT discards IPv4 packets
                 and
                 sends an ICMPv4 "Destination Unreachable, Source Route Failed" (Type 3, Code 5)
                       error message
        to the IPv4 node
       }
   }
```

TP Id	TP_SIIT_T46_IPHDR_11
Test Objective	Verify that the IUT successfully fragments when the packet is a fragment or the DF bit is not set and the packet size is greater than the minimum IPv6 MTU in the network set by the translator configuration function.
Reference	IETF RFC 7915 [1], clause 4.1
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_1_5

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length equal to 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
```

```
                Total Length,
                Identification (Fragment ID),
                Flags, with the DF set to 0
                Fragment Offset,
                Time to Live (TTL), with the value greater than or equal to 2
                Protocol,
                Header Checksum,
                Source Address,
                Destination Address,
           No Options
           UDP Header
           Data
      from the IPv4 node
   }
   then {
      the IUT sends an IPv6 packet containing
          IPv6 basic header containing
              Version, with the value of 6
              Traffic Class, with same value of TOS in IPv4 Header
              Flow Label, with the value of all 0
              Payload Length, with the value which is Total length value from the IPv4
                              header, plus 8 for the Fragment Header, minus the size of the IPv4
                              header and IPv4 options
              Next Header, with the value of 44
              Hop Limit, with the value of TTL minus 1
              Source Address, using IPv4 source address and IPv6 prefix to synthesize
              Destination Address, using IPv4 destination address and IPv6 prefix to synthesize
           Fragment Header containing
              Next Header, with the value copied from IPv4 header to Next Header field except
                           that ICMPv4(1) is changed to ICMPv6(58)
              Reserved,
              Fragment Offset, with the value copied from Fragment Offset of IPv4 header
              Res,
              M flag, with the value copied More Fragment bit of IPv4 header
              Identification, with the value copied the low-order 16 bits of Identification
                              field of IPv4 header and setting high-order 16 bits to zero
           UDP Header,
           Data
      to the IPv6 node
     }
  }
```

5.2.2.2 Translating ICMPv4 Headers to ICMPv6 Headers

TP Id	TP_SIIT_T46_ICMPH_01
Test Objective	Verify that the IUT successfully adjusts the type values of Echo messages (type 8) to 128, and adjusts the ICMP checksum both to take the type change into account and to include the ICMPv6 pseudo-header.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_2_1
Initial Conditions	

with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
}	

Expected Behaviour	

ensure that {
when {
IUT receives an IPv4 packet with the total length less than 1280 bytes containing
IPv4 basic header containing
Version,
Internet Header Length (IHL),
Type of Service (TOS),
Total Length,
Identification (Fragment ID),
Flags,
Fragment Offset,
Time to Live (TTL), with the value greater than or equal to 2
Protocol,
Header Checksum,
Source Address,
Destination Address,
ICMP header containing	

```
                Type, with the value of 8
                Code,
                Checksum,
                Identifier,
                Sequence Number,
                Data
        from the IPv4 node
    }
    then {
        the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             ICMP Header containing
                Type, with the value of 128
                Code,
                Checksum, with value of taking both the type change into account and to include
                          the ICMPv6 pseudo-header
                Identifier,
                Sequence Number,
                Data
        to the IPv6 node
       }
   }
```

TP Id	TP_SIIT_T46_ICMPH_02
Test Objective	Verify that the IUT successfully adjusts the Type values of Echo Reply messages (type 0) to 129 and adjusts the ICMP checksum both to take the type change into account and to include the ICMPv6 pseudo-header.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_2_1

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 0
                  Code,
                  Checksum,
                  Identifier,
                  Sequence Number,
                  Data
        from the IPv4 node
    }
    then {
        the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
```

```
               Payload Length,
               Next Header,
               Hop Limit,
               Source Address,
               Destination Address,
            ICMP Header containing
               Type, with the value of 129
               Code,
               Checksum, with value of taking both the type change into account and to include
                         the ICMPv6 pseudo-header
               Identifier,
               Sequence Number,
               Data
         to the IPv6 node
        }
   }
```

TP Id	TP_SIIT_T46_ICMPH_03
Test Objective	Verify that the IUT successfully drops information Request messages (Type 15).
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_2_2

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 15
                  Code,
                  Checksum,
                  Identifier,
                  Sequence Number
        from the IPv4 node
    }
    then {
        the IUT drops the packet
       }
   }
```

TP Id	TP_SIIT_T46_ICMPH_04
Test Objective	Verify that the IUT successfully drops information Reply messages (Type 16).
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_2_2

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
```

```
                Internet Header Length (IHL),
                Type of Service (TOS),
                Total Length,
                Identification (Fragment ID),
                Flags,
                Fragment Offset,
                Time to Live (TTL), with the value greater than or equal to 2
                Protocol,
                Header Checksum,
                Source Address,
                Destination Address,
           ICMP header containing
                Type, with the value of 16
                Code,
                Checksum,
                Identifier,
                Sequence Number
       from the IPv4 node
    }
    then {
       the IUT drops the packet
      }
  }
```

TP Id	TP_SIIT_T46_ICMPH_05
Test Objective	Verify that the IUT successfully drops Timestamp and Timestamp Reply messages (Type 13).
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_2_3
Initial Conditions	

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
       }
```

Expected Behaviour

```
ensure that {
    when {
       IUT receives an IPv4 packet with the total length less than 1280 bytes containing
            IPv4 basic header containing
                Version,
                Internet Header Length (IHL),
                Type of Service (TOS),
                Total Length,
                Identification (Fragment ID),
                Flags,
                Fragment Offset,
                Time to Live (TTL), with the value greater than or equal to 2
                Protocol,
                Header Checksum,
                Source Address,
                Destination Address,
           ICMP header containing
                Type, with the value of 13
                Code,
                Checksum,
                Identifier,
                Sequence Number,
                Originate Timestamp,
                Receive Timestamp,
                Transmit Timestamp
       from the IPv4 node
    }
    then {
       the IUT drops the packet
      }
  }
```

TP Id	TP_SIIT_T46_ICMPH_06
Test Objective	Verify that the IUT successfully drops Timestamp and Timestamp Reply messages (Type 14).
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_2_3

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 14
                  Code,
                  Checksum,
                  Identifier,
                  Sequence Number,
                  Originate Timestamp,
                  Receive Timestamp,
                  Transmit Timestamp
        from the IPv4 node
    }
    then {
        the IUT drops the packet
       }
   }
```

TP Id	TP_SIIT_T46_ICMPH_07
Test Objective	Verify that the IUT successfully drops Address Mask Request messages (Type 17).
Reference	IETF RFC 7915 [1], clause 4.2, IETF RFC 950 [5]
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_2_4

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 17
                  Code,
                  Checksum,
```

```
                Identifier,
                Sequence Number,
                Address Mask
        from the IPv4 node
    }
    then {
        the IUT drops the packet
       }
   }
```

TP Id	TP_SIIT_T46_ICMPH_08
Test Objective	Verify that the IUT successfully drops Address Mask Reply messages (Type 18).
Reference	IETF RFC 7915 [1], clause 4.2, IETF RFC 950[5]
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_2_4
Initial Conditions	

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 18
                  Code,
                  Checksum,
                  Identifier,
                  Sequence Number,
                  Address Mask
        from the IPv4 node
    }
    then {
        the IUT drops the packet
       }
   }
```

TP Id	TP_SIIT_T46_ICMPH_09
Test Objective	Verify that the IUT successfully drops ICMP Router Advertisement messages (Type 9).
Reference	IETF RFC 7915 [1], clause 4.2, IETF RFC 1256 [6]
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_2_5
Initial Conditions	

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
```

```
                Time to Live (TTL), with the value greater than or equal to 2
                Protocol,
                Header Checksum,
                Source Address,
                Destination Address,
           ICMP header containing
                Type, with the value of 9
                Code,
                Checksum,
                Num Addrs,
                Addrs Entry Size,
                Lifetime,
                Router Address,
                Preference level
      from the IPv4 node
    }
    then {
        the IUT drops the packet
       }
   }
```

TP Id	TP_SIIT_T46_ICMPH_10
Test Objective	Verify that the IUT successfully drops ICMP Router Advertisement messages (Type 10).
Reference	IETF RFC 7915 [1], clause 4.2, IETF RFC 1256 [6]
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_2_6
Initial Conditions	

with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
}	

Expected Behaviour	

ensure that {
when {
IUT receives an IPv4 packet with the total length less than 1280 bytes containing
IPv4 basic header containing
Version,
Internet Header Length (IHL),
Type of Service (TOS),
Total Length,
Identification (Fragment ID),
Flags,
Fragment Offset,
Time to Live (TTL), with the value greater than or equal to 2
Protocol,
Header Checksum,
Source Address,
Destination Address,
ICMP header containing
Type, with the value of 10
Code,
Checksum,
Reserved, with the value of 0
from the IPv4 node
}
then {
the IUT drops the packet
}
}	

TP Id	TP_SIIT_T46_ICMPH_11
Test Objective	Verify that the IUT successfully drops unknown ICMPv4 types messages.
Reference	IETF RFC 7915 [1], clause 4.2, IETF RFC 1256 [6]
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_2_7
Initial Conditions	

with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
}	

Expected Behaviour	
ensure that { when {	

```
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with unknown value
                  Code,
                  Checksum,
                  Reserved
        from the IPv4 node
    }
    then {
        the IUT drops the packet
       }
   }
```

TP Id	TP_SIIT_T46_ICMPH_12
Test Objective	Verify that the IUT successfully translates Destination Unreachable messages (Type 3) with code 0.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_1_1

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 3
                  Code, with the value of 0
                  Checksum,
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             ICMP Header containing
                Type, with the value of 1
```

```
                Code, with the value of 0
                Checksum, with value of taking both the type/code change into account and to
                          include the ICMPv6 pseudo-header
                Unused,
                Data, copied from Internet Header + 64 bits of Original Data Datagram
        to the IPv6 node
       }
   }
```

TP Id	TP_SIIT_T46_ICMPH_13
Test Objective	Verify that the IUT successfully translates Destination Unreachable messages (Type 3) with code 1.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_1_1

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 3
                  Code, with the value of 1
                  Checksum,
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             ICMP Header containing
                Type, with the value of 1
                Code, with the value of 0
                Checksum, with value of taking both the type/code change into account and to
                          include the ICMPv6 pseudo-header
                Unused,
                Data, copied from Internet Header + 64 bits of Original Data Datagram
        to the IPv6 node
       }
   }
```

TP Id	TP_SIIT_T46_ICMPH_14
Test Objective	Verify that the IUT successfully translates Destination Unreachable messages (Type 3) with code 2.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_1_2

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 3
                  Code, with the value of 2
                  Checksum,
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             ICMP Header containing
                Type, with the value of 4
                Code, with the value of 1
                Checksum, with value of taking both the type/code change into account and to
                          include the ICMPv6 pseudo-header
                Pointer, point to the IPv6 Next Header field (6)
                Data, Internet Header + 64 bits of Original Data Datagram
        to the IPv6 node
       }
   }
```

TP Id	TP_SIIT_T46_ICMPH_15
Test Objective	Verify that the IUT successfully translates Destination Unreachable messages (Type 3) with code 3.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_1_3

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
```

```
                Version,
                Internet Header Length (IHL),
                Type of Service (TOS),
                Total Length,
                Identification (Fragment ID),
                Flags,
                Fragment Offset,
                Time to Live (TTL), with the value greater than or equal to 2
                Protocol,
                Header Checksum,
                Source Address,
                Destination Address,
           ICMP header containing
                Type, with the value of 3
                Code, with the value of 3
                Checksum,
                unused,
                Internet Header + 64 bits of Original Data Datagram
      from the IPv4 node
   }
   then {
        the IUT sends an IPv6 packet containing
          IPv6 basic header containing
              Version,
              Traffic Class,
              Flow Label,
              Payload Length,
              Next Header,
              Hop Limit,
              Source Address,
              Destination Address,
           ICMP Header containing
              Type, with the value of 1
              Code, with the value of 4
              Checksum, with value of taking both the type/code change into account and to
                        include the ICMPv6 pseudo-header
              Pointer, point to the IPv6 Next Header field
              Data, copied from Internet Header + 64 bits of Original Data Datagram
      to the IPv6 node
     }
  }
```

TP Id	TP_SIIT_T46_ICMPH_16
Test Objective	Verify that the IUT successfully translates Destination Unreachable messages (Type 3) with code 4.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_1_4_1
Initial Conditions	

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
       }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 3
                  Code, with the value of 4
                  Checksum,
                  unused,
```

```
                Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             ICMP Header containing
                Type, with the value of 2
                Code, with the value of 0
                Checksum, with value of taking both the type/code change into account and to
                          include the ICMPv6 pseudo-header
                MTU, adjusted for the difference between the IPv4 and IPv6 header sizes,
                     but cannot be set to a value smaller than the minimum IPv6 MTU (1280 bytes)
                Data, copy as much of invoking packet as possible without the ICMPv6 packet
                      exceeding the minimum IPv6 MTU
        to the IPv6 node
       }
   }
```

TP Id	TP_SIIT_T46_ICMPH_17
Test Objective	Verify that the IUT successfully translates Destination Unreachable messages (Type 3) with code 4 if the MTU value is zero using the format defined in IETF RFC 4884 [4].
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_1_4_2

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 3
                  Code, with the value of 4
                  Checksum,
                  unused,
                  Length,
                  MTU, with the value of 0
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
```

```
               Destination Address,
            ICMP Header containing
               Type, with the value of 2
               Code, with the value of 0
               Checksum, with value of taking both the type/code change into account and to
                         include the ICMPv6 pseudo-header
               MTU, Use the greatest plateau value (IETF RFC 1191 [3]) that is less than the
                    returned Total Length field, but that is larger than or equal to 1280
               Data, copy as much of invoking packet as possible without the ICMPv6 packet
                     exceeding the minimum IPv6 MTU
        to the IPv6 node
       }
   }
```

TP Id	TP_SIIT_T46_ICMPH_18
Test Objective	Verify that the IUT successfully translates Destination Unreachable messages (Type 3) with code 5.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_1_5

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
       }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 3
                  Code, with the value of 5
                  Checksum,
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             ICMP Header containing
                Type, with the value of 1
                Code, with the value of 0
                Checksum, with value of taking both the type/code change into account and to
                          include the ICMPv6 pseudo-header
                Unused,
                Data, copied from Internet Header + 64 bits of Original Data Datagram
        to the IPv6 node
       }
   }
```

TP Id	TP_SIIT_T46_ICMPH_19
Test Objective	Verify that the IUT successfully translates Destination Unreachable messages (Type 3) with code 6.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_1_6

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 3
                  Code, with the value of 6
                  Checksum,
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             ICMP Header containing
                Type, with the value of 1
                Code, with the value of 0
                Checksum, with value of taking both the type/code change into account and to
                          include the ICMPv6 pseudo-header
                Unused,
                Data, copied from Internet Header + 64 bits of Original Data Datagram
        to the IPv6 node
       }
   }
```

TP Id	TP_SIIT_T46_ICMPH_20
Test Objective	Verify that the IUT successfully translates Destination Unreachable messages (Type 3) with code 7.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_1_6

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
```

```
                Version,
                Internet Header Length (IHL),
                Type of Service (TOS),
                Total Length,
                Identification (Fragment ID),
                Flags,
                Fragment Offset,
                Time to Live (TTL), with the value greater than or equal to 2
                Protocol,
                Header Checksum,
                Source Address,
                Destination Address,
           ICMP header containing
                Type, with the value of 3
                Code, with the value of 7
                Checksum,
                unused,
                Internet Header + 64 bits of Original Data Datagram
      from the IPv4 node
   }
   then {
        the IUT sends an IPv6 packet containing
          IPv6 basic header containing
              Version,
              Traffic Class,
              Flow Label,
              Payload Length,
              Next Header,
              Hop Limit,
              Source Address,
              Destination Address,
           ICMP Header containing
              Type, with the value of 1
              Code, with the value of 0
              Checksum, with value of taking both the type/code change into account and to
                        include the ICMPv6 pseudo-header
              Unused,
              Data, copied from Internet Header + 64 bits of Original Data Datagram
      to the IPv6 node
     }
  }
```

TP Id	TP_SIIT_T46_ICMPH_21
Test Objective	Verify that the IUT successfully translates Destination Unreachable messages (Type 3) with code 8.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_1_6
Initial Conditions	

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
       }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 3
                  Code, with the value of 8
                  Checksum,
                  unused,
```

```
                Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             ICMP Header containing
                Type, with the value of 1
                Code, with the value of 0
                Checksum, with value of taking both the type/code change into account and to
                          include the ICMPv6 pseudo-header
                Unused,
                Data, copied from Internet Header + 64 bits of Original Data Datagram
        to the IPv6 node
       }
  }
```

TP Id	TP_SIIT_T46_ICMPH_22
Test Objective	Verify that the IUT successfully translates Destination Unreachable messages (Type 3) with code 9.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_1_7
Initial Conditions	

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 3
                  Code, with the value of 9
                  Checksum,
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             ICMP Header containing
                Type, with the value of 1
                Code, with the value of 1
```

```
                Checksum, with value of taking both the type/code change into account and to
                          include the ICMPv6 pseudo-header
                Unused,
                Data, copied from Internet Header + 64 bits of Original Data Datagram
        to the IPv6 node
       }
   }
```

TP Id	TP_SIIT_T46_ICMPH_23
Test Objective	Verify that the IUT successfully translates Destination Unreachable messages (Type 3) with code 10.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_1_7

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 3
                  Code, with the value of 10
                  Checksum,
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             ICMP Header containing
                Type, with the value of 1
                Code, with the value of 1
                Checksum, with value of taking both the type/code change into account and to
                          include the ICMPv6 pseudo-header
                Unused,
                Data, copied from Internet Header + 64 bits of Original Data Datagram
        to the IPv6 node
       }
   }
```

TP Id	TP_SIIT_T46_ICMPH_24
Test Objective	Verify that the IUT successfully translates Destination Unreachable messages (Type 3) with code 11.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_1_8

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 3
                  Code, with the value of 11
                  Checksum,
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             ICMP Header containing
                Type, with the value of 1
                Code, with the value of 0
                Checksum, with value of taking both the type/code change into account and to
                          include the ICMPv6 pseudo-header
                Unused,
                Data, copied from Internet Header + 64 bits of Original Data Datagram
        to the IPv6 node
       }
   }
```

TP Id	TP_SIIT_T46_ICMPE_01
Test Objective	Verify that the IUT successfully translates Destination Unreachable messages (Type 3) with code 12.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_1_6

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
```

```
                Version,
                Internet Header Length (IHL),
                Type of Service (TOS),
                Total Length,
                Identification (Fragment ID),
                Flags,
                Fragment Offset,
                Time to Live (TTL), with the value greater than or equal to 2
                Protocol,
                Header Checksum,
                Source Address,
                Destination Address,
           ICMP header containing
                Type, with the value of 3
                Code, with the value of 12
                Checksum,
                unused,
                Internet Header + 64 bits of Original Data Datagram
      from the IPv4 node
   }
   then {
        the IUT sends an IPv6 packet containing
          IPv6 basic header containing
              Version,
              Traffic Class,
              Flow Label,
              Payload Length,
              Next Header,
              Hop Limit,
              Source Address,
              Destination Address,
           ICMP Header containing
              Type, with the value of 1
              Code, with the value of 0
              Checksum, with value of taking both the type/code change into account and to
                        include the ICMPv6 pseudo-header
              Unused,
              Data, copied from Internet Header + 64 bits of Original Data Datagram
      to the IPv6 node
     }
  }
```

TP Id	TP_SIIT_T46_ICMPH_25
Test Objective	Verify that the IUT successfully translates Destination Unreachable messages (Type 3) with code 13.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_1_9
Initial Conditions	

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
       }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 3
                  Code, with the value of 13
                  Checksum,
                  unused,
```

```
                Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             ICMP Header containing
                Type, with the value of 1
                Code, with the value of 1
                Checksum, with value of taking both the type/code change into account and to
                          include the ICMPv6 pseudo-header
                Unused,
                Data, copied from Internet Header + 64 bits of Original Data Datagram
        to the IPv6 node
       }
  }
```

TP Id	TP_SIIT_T46_ICMPH_26
Test Objective	Verify that the IUT successfully translates Destination Unreachable messages (Type 3) with code 14.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_1_9
Initial Conditions	

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
       }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 3
                  Code, with the value of 14
                  Checksum,
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT drops that IPv4 packet
       }
  }
```

TP Id	TP_SIIT_T46_ICMPH_27
Test Objective	Verify that the IUT successfully translates Destination Unreachable messages (Type 3) with code 15.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_1_11

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 3
                  Code, with the value of 15
                  Checksum,
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             ICMP Header containing
                Type, with the value of 1
                Code, with the value of 1
                Checksum, with value of taking both the type/code change into account and to
                          include the ICMPv6 pseudo-header
                Unused,
                Data, copied from Internet Header + 64 bits of Original Data Datagram
        to the IPv6 node
       }
   }
```

TP Id	TP_SIIT_T46_ICMPH_28
Test Objective	Verify that the IUT successfully translates Destination Unreachable messages (Type 3) with other code values (20).
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_1_12

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
```

```
                Version,
                Internet Header Length (IHL),
                Type of Service (TOS),
                Total Length,
                Identification (Fragment ID),
                Flags,
                Fragment Offset,
                Time to Live (TTL), with the value greater than or equal to 2
                Protocol,
                Header Checksum,
                Source Address,
                Destination Address,
           ICMP header containing
                Type, with the value of 3
                Code, with the value of 20
                Checksum,
                unused,
                Internet Header + 64 bits of Original Data Datagram
       from the IPv4 node
   }
   then {
        the IUT drops that IPv4 packet
      }
  }
```

TP Id	TP_SIIT_T46_ICMPH_29
Test Objective	Verify that the IUT successfully translates Redirect messages (Type 5).
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_2
Initial Conditions	

with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
}	

Expected Behaviour	

ensure that {
when {
IUT receives an IPv4 packet with the total length less than 1280 bytes containing
IPv4 basic header containing
Version,
Internet Header Length (IHL),
Type of Service (TOS),
Total Length,
Identification (Fragment ID),
Flags,
Fragment Offset,
Time to Live (TTL), with the value greater than or equal to 2
Protocol,
Header Checksum,
Source Address,
Destination Address,
ICMP header containing
Type, with the value of 5
Code,
Checksum,
Gateway Internet Address,
Internet Header + 64 bits of Original Data Datagram
from the IPv4 node
}
then {
the IUT drops that IPv4 packet
}
}	

TP Id	TP_SIIT_T46_ICMPH_30
Test Objective	Verify that the IUT successfully translates Alternative Host Address messages (Type 6). NOTE: There is no public information about this type.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_3

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 6,
                  Code,
                  Checksum,
                  Unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT drops that IPv4 packet
       }
   }
```

TP Id	TP_SIIT_T46_ICMPH_31
Test Objective	Verify that the IUT successfully translates Source Quench messages (Type 4).
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_4

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 4
                  Code,
                  Checksum,
                  Unused,
                  Internet Header + 64 bits of Original Data Datagram
```

```
        from the IPv4 node
    }
    then {
         the IUT drops that IPv4 packet
       }
  }
```

TP Id	TP_SIIT_T46_ICMPH_32
Test Objective	Verify that the IUT successfully translates Time Exceeded messages (Type 11).
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_5

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 11
                  Code,
                  Checksum,
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             ICMP Header containing
                Type, with the value of 3
                Code,
                Checksum, with value of taking both the type/code change into account and to
                          include the ICMPv6 pseudo-header
                Unused,
                Data, copied from Internet Header + 64 bits of Original Data Datagram
        to the IPv6 node
       }
  }
```

TP Id	TP_SIIT_T46_ICMPH_33
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 0 and pointer 0.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_1

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 12
                  Code, with the value of 0
                  Checksum,
                  Pointer, with the value of 0
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             ICMP Header containing
                Type, with the value of 4
                Code, with the value of 0
                Checksum, with value of taking both the type/code change into account and to
                          include the ICMPv6 pseudo-header
                Pointer, with the value of 0
                Data, copied from Internet Header + 64 bits of Original Data Datagram
        to the IPv6 node
       }
   }
```

TP Id	TP_SIIT_T46_ICMPH_34
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 0 and pointer 1.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_1

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
```

```
            IPv4 basic header containing
                 Version,
                 Internet Header Length (IHL),
                 Type of Service (TOS),
                 Total Length,
                 Identification (Fragment ID),
                 Flags,
                 Fragment Offset,
                 Time to Live (TTL), with the value greater than or equal to 2
                 Protocol,
                 Header Checksum,
                 Source Address,
                 Destination Address,
            ICMP header containing
                 Type, with the value of 12
                 Code, with the value of 0
                 Pointer, with the value of 1
                 Checksum,
                 unused,
                 Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
            ICMP Header containing
                Type, with the value of 4
                Code, with the value of 0
                Checksum, with value of taking both the type/code change into account and to
                          include the ICMPv6 pseudo-header
                Pointer, with the value of 1
                Data, copied from Internet Header + 64 bits of Original Data Datagram
        to the IPv6 node
       }
  }
```

TP Id	TP_SIIT_T46_ICMPH_35
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 0 and pointer 2.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_1
Initial Conditions	

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
       }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 12
                  Code, with the value of 0
```

```
                Checksum,
                Pointer, with the value of 2
                unused,
                Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             ICMP Header containing
                Type, with the value of 4
                Code, with the value of 0
                Checksum, with value of taking both the type/code change into account and to
                          include the ICMPv6 pseudo-header
                Pointer, with the value of 4
                Data, copied from Internet Header + 64 bits of Original Data Datagram
        to the IPv6 node
       }
   }
```

TP Id	TP_SIIT_T46_ICMPH_36
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 0 and pointer 3.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_1
Initial Conditions	

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 12
                  Code, with the value of 0
                  Checksum,
                  Pointer, with the value of 3
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
```

```
               Destination Address,
            ICMP Header containing
               Type, with the value of 4
               Code, with the value of 0
               Checksum, with value of taking both the type/code change into account and to
                         include the ICMPv6 pseudo-header
               Pointer, with the value of 4
               Data, copied from Internet Header + 64 bits of Original Data Datagram
         to the IPv6 node
        }
   }
```

TP Id	TP_SIIT_T46_ICMPH_37
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 0 and pointer 4.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_1
Initial Conditions	

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 12
                  Code, with the value of 0
                  Checksum,
                  Pointer, with the value of 4
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT drops that IPv4 packet
       }
   }
```

TP Id	TP_SIIT_T46_ICMPH_38
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 0 and pointer 5.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_1
Initial Conditions	

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
```

```
                Type of Service (TOS),
                Total Length,
                Identification (Fragment ID),
                Flags,
                Fragment Offset,
                Time to Live (TTL), with the value greater than or equal to 2
                Protocol,
                Header Checksum,
                Source Address,
                Destination Address,
           ICMP header containing
                Type, with the value of 12
                Code, with the value of 0
                Checksum,
                Pointer, with the value of 5
                unused,
                Internet Header + 64 bits of Original Data Datagram
       from the IPv4 node
   }
   then {
        the IUT drops that IPv4 packet
      }
  }
```

TP Id	TP_SIIT_T46_ICMPH_39
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 0 and pointer 6.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_1

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
       }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 12
                  Code, with the value of 0
                  Checksum,
                  Pointer, with the value of 6
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT drops that IPv4 packet
       }
  }
```

TP Id	TP_SIIT_T46_ICMPH_40
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 0 and pointer 7.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_1
Initial Conditions	

with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
}	

Expected Behaviour	

ensure that {
when {
IUT receives an IPv4 packet with the total length less than 1280 bytes containing
IPv4 basic header containing
Version,
Internet Header Length (IHL),
Type of Service (TOS),
Total Length,
Identification (Fragment ID),
Flags,
Fragment Offset,
Time to Live (TTL), with the value greater than or equal to 2
Protocol,
Header Checksum,
Source Address,
Destination Address,
ICMP header containing
Type, with the value of 12
Code, with the value of 0
Checksum,
Pointer, with the value of 7
unused,
Internet Header + 64 bits of Original Data Datagram
from the IPv4 node
}
then {
the IUT drops that IPv4 packet
}
}	

TP Id	TP_SIIT_T46_ICMPH_41
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 0 and pointer 8.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_1
Initial Conditions	

with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
}	

Expected Behaviour	

ensure that {
when {
IUT receives an IPv4 packet with the total length less than 1280 bytes containing
IPv4 basic header containing
Version,
Internet Header Length (IHL),
Type of Service (TOS),
Total Length,
Identification (Fragment ID),
Flags,
Fragment Offset,
Time to Live (TTL), with the value greater than or equal to 2
Protocol,
Header Checksum,
Source Address,
Destination Address,
ICMP header containing
Type, with the value of 12
Code, with the value of 0
Checksum,	

```
                Pointer, with the value of 8
                unused,
                Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             ICMP Header containing
                Type, with the value of 4
                Code, with the value of 0
                Checksum, with value of taking both the type/code change into account and to
                          include the ICMPv6 pseudo-header
                Pointer, with the value of 7
                Data, copied from Internet Header + 64 bits of Original Data Datagram
        to the IPv6 node
       }
   }
```

TP Id	TP_SIIT_T46_ICMPH_42
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 0 and pointer 9.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_1
Initial Conditions	

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 12
                  Code, with the value of 0
                  Checksum,
                  Pointer, with the value of 9
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
```

```
            ICMP Header containing
               Type, with the value of 4
               Code, with the value of 0
               Checksum, with value of taking both the type/code change into account and to
                         include the ICMPv6 pseudo-header
               Pointer, with the value of 8
               Data, copied from Internet Header + 64 bits of Original Data Datagram
         to the IPv6 node
        }
   }
```

TP Id	TP_SIIT_T46_ICMPH_43
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 0 and pointer 10.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_1

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 12
                  Code, with the value of 0
                  Checksum,
                  Pointer, with the value of 10
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT drops that IPv4 packet
       }
   }
```

TP Id	TP_SIIT_T46_ICMPH_44
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 0 and pointer 11.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_1

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
```

```
                Total Length,
                Identification (Fragment ID),
                Flags,
                Fragment Offset,
                Time to Live (TTL), with the value greater than or equal to 2
                Protocol,
                Header Checksum,
                Source Address,
                Destination Address,
           ICMP header containing
                Type, with the value of 12
                Code, with the value of 0
                Checksum,
                Pointer, with the value of 11
                unused,
                Internet Header + 64 bits of Original Data Datagram
      from the IPv4 node
   }
   then {
        the IUT drops that IPv4 packet
      }
  }
```

TP Id	TP_SIIT_T46_ICMPH_45
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 0 and pointer 12.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_1

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
       }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 12
                  Code, with the value of 0
                  Checksum,
                  Pointer, with the value of 12
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             ICMP Header containing
                Type, with the value of 4
                Code, with the value of 0
                Checksum, with value of taking both the type/code change into account and to
```

```
                        include the ICMPv6 pseudo-header
                Pointer, with the value of 8
                Data, copied from Internet Header + 64 bits of Original Data Datagram
        to the IPv6 node
       }
   }
```

TP Id	TP_SIIT_T46_ICMPH_46
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 0 and pointer 13.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_1

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
       }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 12
                  Code, with the value of 0
                  Checksum,
                  Pointer, with the value of 13
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             ICMP Header containing
                Type, with the value of 4
                Code, with the value of 0
                Checksum, with value of taking both the type/code change into account and to
                          include the ICMPv6 pseudo-header
                Pointer, with the value of 8
                Data, copied from Internet Header + 64 bits of Original Data Datagram
        to the IPv6 node
       }
   }
```

TP Id	TP_SIIT_T46_ICMPH_47
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 0 and pointer 14.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_1

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 12
                  Code, with the value of 0
                  Checksum,
                  Pointer, with the value of 14
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             ICMP Header containing
                Type, with the value of 4
                Code, with the value of 0
                Checksum, with value of taking both the type/code change into account and to
                          include the ICMPv6 pseudo-header
                Pointer, with the value of 8
                Data, copied from Internet Header + 64 bits of Original Data Datagram
        to the IPv6 node
       }
   }
```

TP Id	TP_SIIT_T46_ICMPH_48
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 0 and pointer 15.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_1

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
```

```
            IPv4 basic header containing
                 Version,
                 Internet Header Length (IHL),
                 Type of Service (TOS),
                 Total Length,
                 Identification (Fragment ID),
                 Flags,
                 Fragment Offset,
                 Time to Live (TTL), with the value greater than or equal to 2
                 Protocol,
                 Header Checksum,
                 Source Address,
                 Destination Address,
            ICMP header containing
                 Type, with the value of 12
                 Code, with the value of 0
                 Checksum,
                 Pointer, with the value of 15
                 unused,
                 Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
            ICMP Header containing
                Type, with the value of 4
                Code, with the value of 0
                Checksum, with value of taking both the type/code change into account and to
                          include the ICMPv6 pseudo-header
                Pointer, with the value of 8
                Data, copied from Internet Header + 64 bits of Original Data Datagram
        to the IPv6 node
       }
  }
```

TP Id	TP_SIIT_T46_ICMPH_49
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 0 and pointer 16.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_1
Initial Conditions	

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
       }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 12
                  Code, with the value of 0
```

```
                Checksum,
                Pointer, with the value of 16
                unused,
                Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             ICMP Header containing
                Type, with the value of 4
                Code, with the value of 0
                Checksum, with value of taking both the type/code change into account and to
                          include the ICMPv6 pseudo-header
                Pointer, with the value of 24
                Data, copied from Internet Header + 64 bits of Original Data Datagram
        to the IPv6 node
       }
  }
```

TP Id	TP_SIIT_T46_ICMPH_50
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 0 and pointer 17.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_1
Initial Conditions	

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 12
                  Code, with the value of 0
                  Checksum,
                  Pointer, with the value of 17
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
```

```
               Destination Address,
            ICMP Header containing
               Type, with the value of 4
               Code, with the value of 0
               Checksum, with value of taking both the type/code change into account and to
                         include the ICMPv6 pseudo-header
               Pointer, with the value of 24
               Data, copied from Internet Header + 64 bits of Original Data Datagram
        to the IPv6 node
       }
   }
```

TP Id	TP_SIIT_T46_ICMPH_51
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 0 and pointer 18.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_1

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
       }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 12
                  Code, with the value of 0
                  Checksum,
                  Pointer, with the value of 18
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             ICMP Header containing
                Type, with the value of 4
                Code, with the value of 0
                Checksum, with value of taking both the type/code change into account and to
                          include the ICMPv6 pseudo-header
                Pointer, with the value of 24
                Data, copied from Internet Header + 64 bits of Original Data Datagram
        to the IPv6 node
       }
   }
```

TP Id	TP_SIIT_T46_ICMPH_52
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 0 and pointer 19.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_1

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 12
                  Code, with the value of 0
                  Checksum,
                  Pointer, with the value of 19
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             ICMP Header containing
                Type, with the value of 4
                Code, with the value of 0
                Checksum, with value of taking both the type/code change into account and to
                          include the ICMPv6 pseudo-header
                Pointer, with the value of 24
                Data, copied from Internet Header + 64 bits of Original Data Datagram
        to the IPv6 node
       }
   }
```

TP Id	TP_SIIT_T46_ICMPH_53
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 1.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_2

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
```

```
                Version,
                Internet Header Length (IHL),
                Type of Service (TOS),
                Total Length,
                Identification (Fragment ID),
                Flags,
                Fragment Offset,
                Time to Live (TTL), with the value greater than or equal to 2
                Protocol,
                Header Checksum,
                Source Address,
                Destination Address,
           ICMP header containing
                Type, with the value of 12
                Code, with the value of 1
                Checksum,
                Pointer,
                unused,
                Internet Header + 64 bits of Original Data Datagram
      from the IPv4 node
   }
   then {
        the IUT drops that IPv4 packet
      }
  }
```

TP Id	TP_SIIT_T46_ICMPH_54
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 2 and pointer 0.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_3

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
       }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 12
                  Code, with the value of 2
                  Checksum,
                  Pointer, with the value of 0
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             ICMP Header containing
```

```
                Type, with the value of 4
                Code, with the value of 0
                Checksum, with value of taking both the type/code change into account and to
                          include the ICMPv6 pseudo-header
                Pointer, with the value of 0
                Data, copied from Internet Header + 64 bits of Original Data Datagram
        to the IPv6 node
       }
   }
```

TP Id	TP_SIIT_T46_ICMPH_55
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 2 and pointer 1.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_3

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 12
                  Code, with the value of 2
                  Checksum,
                  Pointer, with the value of 1
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             ICMP Header containing
                Type, with the value of 4
                Code, with the value of 0
                Checksum, with value of taking both the type/code change into account and to
                          include the ICMPv6 pseudo-header
                Pointer, with the value of 1
                Data, copied from Internet Header + 64 bits of Original Data Datagram
        to the IPv6 node
       }
   }
```

TP Id	TP_SIIT_T46_ICMPH_56
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 2 and pointer 2.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_3

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 12
                  Code, with the value of 2
                  Checksum,
                  Pointer, with the value of 2
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             ICMP Header containing
                Type, with the value of 4
                Code, with the value of 0
                Checksum, with value of taking both the type/code change into account and to
                          include the ICMPv6 pseudo-header
                Pointer, with the value of 4
                Data, copied from Internet Header + 64 bits of Original Data Datagram
        to the IPv6 node
       }
   }
```

TP Id	TP_SIIT_T46_ICMPH_57
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 2 and pointer 3.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_3

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
```

```
            IPv4 basic header containing
                 Version,
                 Internet Header Length (IHL),
                 Type of Service (TOS),
                 Total Length,
                 Identification (Fragment ID),
                 Flags,
                 Fragment Offset,
                 Time to Live (TTL), with the value greater than or equal to 2
                 Protocol,
                 Header Checksum,
                 Source Address,
                 Destination Address,
            ICMP header containing
                 Type, with the value of 12
                 Code, with the value of 2
                 Checksum,
                 Pointer, with the value of 3
                 unused,
                 Internet Header + 64 bits of Original Data Datagram
       from the IPv4 node
   }
   then {
        the IUT sends an IPv6 packet containing
           IPv6 basic header containing
               Version,
               Traffic Class,
               Flow Label,
               Payload Length,
               Next Header,
               Hop Limit,
               Source Address,
               Destination Address,
            ICMP Header containing
               Type, with the value of 4
               Code, with the value of 0
               Checksum, with value of taking both the type/code change into account and to
                         include the ICMPv6 pseudo-header
               Pointer, with the value of 4
               Data, copied from Internet Header + 64 bits of Original Data Datagram
        to the IPv6 node
       }
  }
```

TP Id	TP_SIIT_T46_ICMPH_58
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 2 and pointer 4.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_3
Initial Conditions	

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
       }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 12
                  Code, with the value of 2
```

```
                Checksum,
                Pointer, with the value of 4
                unused,
                Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT drops that IPv4 packet
        }
   }
```

TP Id	TP_SIIT_T46_ICMPH_59
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 2 and pointer 5.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_3

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 12
                  Code, with the value of 2
                  Checksum,
                  Pointer, with the value of 5
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT drops that IPv4 packet
        }
   }
```

TP Id	TP_SIIT_T46_ICMPH_60
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 2 and pointer 6.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_3

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
```

```
                Total Length,
                Identification (Fragment ID),
                Flags,
                Fragment Offset,
                Time to Live (TTL), with the value greater than or equal to 2
                Protocol,
                Header Checksum,
                Source Address,
                Destination Address,
           ICMP header containing
                Type, with the value of 12
                Code, with the value of 2
                Checksum,
                Pointer, with the value of 6
                unused,
                Internet Header + 64 bits of Original Data Datagram
       from the IPv4 node
   }
   then {
        the IUT drops that IPv4 packet
      }
  }
```

TP Id	TP_SIIT_T46_ICMPH_61
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 2 and pointer 7.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_3

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
       }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
           IPv4 basic header containing
                Version,
                Internet Header Length (IHL),
                Type of Service (TOS),
                Total Length,
                Identification (Fragment ID),
                Flags,
                Fragment Offset,
                Time to Live (TTL), with the value greater than or equal to 2
                Protocol,
                Header Checksum,
                Source Address,
                Destination Address,
           ICMP header containing
                Type, with the value of 12
                Code, with the value of 2
                Checksum,
                Pointer, with the value of 7
                unused,
                Internet Header + 64 bits of Original Data Datagram
       from the IPv4 node
   }
   then {
        the IUT drops that IPv4 packet
      }
  }
```

TP Id	TP_SIIT_T46_ICMPH_62
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 2 and pointer 8.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_3

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 12
                  Code, with the value of 2
                  Checksum,
                  Pointer, with the value of 8
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             ICMP Header containing
                Type, with the value of 4
                Code, with the value of 0
                Checksum, with value of taking both the type/code change into account and to
                          include the ICMPv6 pseudo-header
                Pointer, with the value of 7
                Data, copied from Internet Header + 64 bits of Original Data Datagram
        to the IPv6 node
       }
   }
```

TP Id	TP_SIIT_T46_ICMPH_63
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 2 and pointer 9.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_3

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
```

```
            IPv4 basic header containing
                 Version,
                 Internet Header Length (IHL),
                 Type of Service (TOS),
                 Total Length,
                 Identification (Fragment ID),
                 Flags,
                 Fragment Offset,
                 Time to Live (TTL), with the value greater than or equal to 2
                 Protocol,
                 Header Checksum,
                 Source Address,
                 Destination Address,
            ICMP header containing
                 Type, with the value of 12
                 Code, with the value of 2
                 Checksum,
                 Pointer, with the value of 9
                 unused,
                 Internet Header + 64 bits of Original Data Datagram
       from the IPv4 node
   }
   then {
        the IUT sends an IPv6 packet containing
           IPv6 basic header containing
               Version,
               Traffic Class,
               Flow Label,
               Payload Length,
               Next Header,
               Hop Limit,
               Source Address,
               Destination Address,
            ICMP Header containing
               Type, with the value of 4
               Code, with the value of 0
               Checksum, with value of taking both the type/code change into account and to
                         include the ICMPv6 pseudo-header
               Pointer, with the value of 8
               Data, copied from Internet Header + 64 bits of Original Data Datagram
       to the IPv6 node
      }
  }
```

TP Id	TP_SIIT_T46_ICMPH_64
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 2 and pointer 10.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_3
Initial Conditions	

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
       }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 12
                  Code, with the value of 2
```

```
                Checksum,
                Pointer, with the value of 10
                unused,
                Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT drops that IPv4 packet
        }
   }
```

TP Id	TP_SIIT_T46_ICMPH_65
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 2 and pointer 11.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_3

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 12
                  Code, with the value of 2
                  Checksum,
                  Pointer, with the value of 11
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT drops that IPv4 packet
        }
   }
```

TP Id	TP_SIIT_T46_ICMPH_66
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 2 and pointer 12.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_3

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
```

```
                Total Length,
                Identification (Fragment ID),
                Flags,
                Fragment Offset,
                Time to Live (TTL), with the value greater than or equal to 2
                Protocol,
                Header Checksum,
                Source Address,
                Destination Address,
           ICMP header containing
                Type, with the value of 12
                Code, with the value of 2
                Checksum,
                Pointer, with the value of 12
                unused,
                Internet Header + 64 bits of Original Data Datagram
      from the IPv4 node
   }
   then {
        the IUT sends an IPv6 packet containing
          IPv6 basic header containing
              Version,
              Traffic Class,
              Flow Label,
              Payload Length,
              Next Header,
              Hop Limit,
              Source Address,
              Destination Address,
           ICMP Header containing
              Type, with the value of 4
              Code, with the value of 0
              Checksum, with value of taking both the type/code change into account and to
                        include the ICMPv6 pseudo-header
              Pointer, with the value of 8
              Data, copied from Internet Header + 64 bits of Original Data Datagram
      to the IPv6 node
     }
  }
```

TP Id	TP_SIIT_T46_ICMPH_67
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 2 and pointer 13.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_3
Initial Conditions	

with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
}	

Expected Behaviour	

ensure that {
when {
IUT receives an IPv4 packet with the total length less than 1280 bytes containing
IPv4 basic header containing
Version,
Internet Header Length (IHL),
Type of Service (TOS),
Total Length,
Identification (Fragment ID),
Flags,
Fragment Offset,
Time to Live (TTL), with the value greater than or equal to 2
Protocol,
Header Checksum,
Source Address,
Destination Address,
ICMP header containing
Type, with the value of 12
Code, with the value of 2
Checksum,
Pointer, with the value of 13
unused,
Internet Header + 64 bits of Original Data Datagram	

```
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             ICMP Header containing
                Type, with the value of 4
                Code, with the value of 0
                Checksum, with value of taking both the type/code change into account and to
                          include the ICMPv6 pseudo-header
                Pointer, with the value of 8
                Data, copied from Internet Header + 64 bits of Original Data Datagram
        to the IPv6 node
       }
   }
```

TP Id	TP_SIIT_T46_ICMPH_68
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 2 and pointer 14.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_3

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 12
                  Code, with the value of 2
                  Checksum,
                  Pointer, with the value of 14
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             ICMP Header containing
                Type, with the value of 4
                Code, with the value of 0
```

```
                Checksum, with value of taking both the type/code change into account and to
                          include the ICMPv6 pseudo-header
                Pointer, with the value of 8
                Data, copied from Internet Header + 64 bits of Original Data Datagram
        to the IPv6 node
       }
   }
```

TP Id	TP_SIIT_T46_ICMPH_69
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 2 and pointer 15.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_3

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 12
                  Code, with the value of 2
                  Checksum,
                  Pointer, with the value of 15
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             ICMP Header containing
                Type, with the value of 4
                Code, with the value of 0
                Checksum, with value of taking both the type/code change into account and to
                          include the ICMPv6 pseudo-header
                Pointer, with the value of 8
                Data, copied from Internet Header + 64 bits of Original Data Datagram
        to the IPv6 node
       }
   }
```

TP Id	TP_SIIT_T46_ICMPH_70
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 3 and pointer 16.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_2

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 12
                  Code, with the value of 2
                  Checksum,
                  Pointer, with the value of 16
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             ICMP Header containing
                Type, with the value of 4
                Code, with the value of 0
                Checksum, with value of taking both the type/code change into account and to
                          include the ICMPv6 pseudo-header
                Pointer, with the value of 24
                Data, copied from Internet Header + 64 bits of Original Data Datagram
        to the IPv6 node
       }
   }
```

TP Id	TP_SIIT_T46_ICMPH_71
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 2 and pointer 17.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_3

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
```

```
            IPv4 basic header containing
                 Version,
                 Internet Header Length (IHL),
                 Type of Service (TOS),
                 Total Length,
                 Identification (Fragment ID),
                 Flags,
                 Fragment Offset,
                 Time to Live (TTL), with the value greater than or equal to 2
                 Protocol,
                 Header Checksum,
                 Source Address,
                 Destination Address,
            ICMP header containing
                 Type, with the value of 12
                 Code, with the value of 2
                 Checksum,
                 Pointer, with the value of 17
                 unused,
                 Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
            ICMP Header containing
                Type, with the value of 4
                Code, with the value of 0
                Checksum, with value of taking both the type/code change into account and to
                          include the ICMPv6 pseudo-header
                Pointer, with the value of 24
                Data, copied from Internet Header + 64 bits of Original Data Datagram
        to the IPv6 node
       }
  }
```

TP Id	TP_SIIT_T46_ICMPH_72
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 2 and pointer 18.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_3
Initial Conditions	

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
       }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 12
                  Code, with the value of 2
```

```
                Checksum,
                Pointer, with the value of 18
                unused,
                Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             ICMP Header containing
                Type, with the value of 4
                Code, with the value of 0
                Checksum, with value of taking both the type/code change into account and to
                          include the ICMPv6 pseudo-header
                Pointer, with the value of 24
                Data, copied from Internet Header + 64 bits of Original Data Datagram
        to the IPv6 node
       }
   }
```

TP Id	TP_SIIT_T46_ICMPH_73
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code 2 and pointer 19.
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_3
Initial Conditions	

with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
}	

Expected Behaviour	

ensure that {
when {
IUT receives an IPv4 packet with the total length less than 1280 bytes containing
IPv4 basic header containing
Version,
Internet Header Length (IHL),
Type of Service (TOS),
Total Length,
Identification (Fragment ID),
Flags,
Fragment Offset,
Time to Live (TTL), with the value greater than or equal to 2
Protocol,
Header Checksum,
Source Address,
Destination Address,
ICMP header containing
Type, with the value of 12
Code, with the value of 2
Checksum,
Pointer, with the value of 19
unused,
Internet Header + 64 bits of Original Data Datagram
from the IPv4 node
}
then {
the IUT sends an IPv6 packet containing
IPv6 basic header containing
Version,
Traffic Class,
Flow Label,
Payload Length,
Next Header,
Hop Limit,
Source Address,	

```
               Destination Address,
            ICMP Header containing
               Type, with the value of 4
               Code, with the value of 0
               Checksum, with value of taking both the type/code change into account and to
                         include the ICMPv6 pseudo-header
               Pointer, with the value of 24
               Data, copied from Internet Header + 64 bits of Original Data Datagram
         to the IPv6 node
        }
   }
```

TP Id	TP_SIIT_T46_ICMPH_74
Test Objective	Verify that the IUT successfully translates Parameter Problem messages (Type 12) with code values (10).
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_6_4

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 12
                  Code, with the value of 10
                  Checksum,
                  Pointer,
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
    }
    then {
         the IUT drops that IPv4 packet
        }
   }
```

TP Id	TP_SIIT_T46_ICMPH_75
Test Objective	Verify that the IUT successfully truncates the extension if the ICMPv4 Extension exceeds the maximum size of an ICMPv6 messages (1 280 bytes).
Reference	IETF RFC 7915 [1], clause 4.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_2_3_7

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
```

```
                Type of Service (TOS),
                Total Length,
                Identification (Fragment ID),
                Flags,
                Fragment Offset,
                Time to Live (TTL), with the value greater than or equal to 2
                Protocol,
                Header Checksum,
                Source Address,
                Destination Address,
           ICMP header containing
                Type, with the value of 12
                Code, with the value of 10
                Checksum,
                Pointer,
                unused,
                Internet Header + 64 bits of Original Data Datagram
       from the IPv4 node
   }
   then {
        the IUT drops that IPv4 packet
      }
  }
```

5.2.2.3 Translating ICMPv4 Error Messages into ICMPv6

TP Id	TP_SIIT_T46_ ICMPE_01
Test Objective	Verify that the IUT successfully translates IPv4 packet inside the ICMPv4 .
Reference	IETF RFC 7915 [1], clause 4.3
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_3_1, 1_3_2

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
       }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 3
                  Code, with the value of 0
                  Checksum,
                  Pointer,
                  unused,
                  Internet Header + 64 bits of Original Data Datagram containing
                         IPv4 basic header containing
                              Version,
                              Internet Header Length (IHL),
                              Type of Service (TOS),
                              Total Length,
                              Identification (Fragment ID),
                              Flags,
                              Fragment Offset,
                              Time to Live (TTL), with the value greater than or equal to 2
                              Protocol,
                              Header Checksum,
                              Source Address,
                              Destination Address,
                         Data Datagram
```

```
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length, with value of taking into account the length change in Data
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             ICMP Header containing
                Type, with the value of 1
                Code, with the value of 0
                Checksum, with value of taking both the type/code change into account and to
                          include the ICMPv6 pseudo-header
                Unused,
                Data containing
                     IPv6 basic header containing
                          Version,
                          Traffic Class,
                          Flow Label,
                          Payload Length,
                          Next Header,
                          Hop Limit,
                          Source Address, using IPv4 source address and IPv6 prefix to synthesize
                          Destination Address, using IPv4 destination address and IPv6 prefix to
                                               synthesize
                     Data, copied from Data Datagram
        to the IPv6 node
       }
  }
```

TP Id	TP_SIIT_T46_ICMPE_02
Test Objective	Verify that the IUT successfully translates the outer IP headers at the first embedded header.
Reference	IETF RFC 7915 [1], clause 4.3
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_3_3
Initial Conditions	

with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
}	

Expected Behaviour	

ensure that {
when {
IUT receives an IPv4 packet with the total length less than 1280 bytes containing
IPv4 basic header containing
Version,
Internet Header Length (IHL),
Type of Service (TOS),
Total Length,
Identification (Fragment ID),
Flags,
Fragment Offset,
Time to Live (TTL), with the value greater than or equal to 2
Protocol,
Header Checksum,
Source Address,
Destination Address,
ICMP header containing
Type, with the value of 3
Code, with the value of 0
Checksum,
Pointer,
unused,
Internet Header + 64 bits of Original Data Datagram containing
IPv4 basic header containing
Version,
Internet Header Length (IHL),
Type of Service (TOS),
Total Length,
Identification (Fragment ID),	

```
                              Flags,
                              Fragment Offset,
                              Time to Live (TTL), with the value greater than or equal to 2
                              Protocol,
                              Header Checksum,
                              Source Address,
                              Destination Address,
                         Data Datagram, without embedded IP headers
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length, with value of taking into account the length change in Data
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             ICMP Header containing
                Type, with the value of 1
                Code, with the value of 0
                Checksum, with value of taking both the type/code change into account and to
                          include the ICMPv6 pseudo-header
                Unused,
                Data containing
                     IPv6 basic header containing
                          Version,
                          Traffic Class,
                          Flow Label,
                          Payload Length,
                          Next Header,
                          Hop Limit,
                          Source Address, using IPv4 source address and IPv6 prefix to synthesize
                          Destination Address, using IPv4 destination address and IPv6 prefix to
                                               synthesize
                     Data, copied from Data Datagram
        to the IPv6 node
       }
   }
```

TP Id	TP_SIIT_T46_ICMPE_03
Test Objective	Verify that the IUT successfully drops the outer IP headers at the first embedded header if it contains more embedded headers.
Reference	IETF RFC 7915 [1], clause 4.3
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_3_3
Initial Conditions	

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 3
                  Code, with the value of 0
                  Checksum,
```

```
                Pointer,
                unused,
                Internet Header + 64 bits of Original Data Datagram containing
                       IPv4 basic header containing
                            Version,
                            Internet Header Length (IHL),
                            Type of Service (TOS),
                            Total Length,
                            Identification (Fragment ID),
                            Flags,
                            Fragment Offset,
                            Time to Live (TTL), with the value greater than or equal to 2
                            Protocol,
                            Header Checksum,
                            Source Address,
                            Destination Address,
                       Data Datagram, with embedded IP headers
        from the IPv4 node
    }
    then {
         the IUT drops that IPv4 packet
       }
   }
```

5.2.2.4 Generation of ICMPv4 Error Message

TP Id	TP_SIIT_T46_GE_01
Test Objective	Verify that the IUT sends back an ICMPv4 error message to the original sender of the packet if the IPv4 packet is discarded.
Reference	IETF RFC 7915 [1], clause 4.4
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_4_1, 1_4_2

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
the IUT is configured to the access of IPv4 node is denied
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             Data
        from the IPv4 node
    }
    then {
        IUT discards that IPv4 packet and
             sends an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL),
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address, with the IPv4 address of that IPv4 packet
```

```
            ICMP header containing
                 Type, with the value of 3
                 Code, with the value of 13
                 Checksum,
                 unused,
                 Internet Header + 64 bits of Original Data Datagram
       to the IPv4 node
      }
  }
```

5.2.2.5 Transport-Layer Header Translation

TP Id	TP_SIIT_T46_TLH_01
Test Objective	Verify that IUT recalculates and updates TCP headers when the address translation algorithm is not checksum neutral.
Reference	IETF RFC 7915 [1], clause 4.5
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_5_1

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
       }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             TCP header containing
                 Source Port,
                 Destination Port,
                 Sequence Number,
                 Acknowledge Number,
                 Data Offset,
                 Reserved,
                 Control Bits,
                 Window,
                 Checksum,
                 Urgent Point,
                 Options,
              Data
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             TCP header containing
                 Source Port,
                 Destination Port,
                 Sequence Number,
                 Acknowledge Number,
                 Data Offset,
                 Reserved,
                 Control Bits,
```

```
                Window,
                Checksum, with the value taking into account IP addresses change
                Urgent Point,
                Options,
              Data
        to the IPv6 node
       }
   }
```

TP Id	TP_SIIT_T46_TLH_02
Test Objective	Verify that IUT recalculates and updates UDP headers when the checksum value is not zero and the address translation algorithm is not checksum neutral.
Reference	IETF RFC 7915 [1], clause 4.5
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_5_1
Initial Conditions	

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             UDP header containing
                 Source Port,
                 Destination Port,
                 Length,
                 Checksum, with the value greater than 0
              Data
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             UDP header containing
                  Source Port,
                  Destination Port,
                  Sequence Number,
                  Checksum, with the value taking into account IP addresses change
             Data
        to the IPv6 node
       }
   }
```

TP Id	TP_SIIT_T46_TLH_03
Test Objective	Verify that IUT recalculates and updates UDP headers when the checksum value is zero and the address translation algorithm is not checksum neutral.
Reference	IETF RFC 7915 [1], clause 4.5
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_5_1

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             UDP header containing
                 Source Port,
                 Destination Port,
                 Length,
                 Checksum, with the value equal to 0
              Data
        from the IPv4 node
    }
    then {
         the IUT sends an IPv6 packet containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             UDP header containing
                 Source Port,
                 Destination Port,
                 Sequence Number,
                 Checksum, calculate the value taking into account IP addresses change
             Data
         to the IPv6 node
                 or drops that IPv4 packet
       }
   }
```

5.2.2.6 Knowing When to Translate

TP Id	TP_SIIT_T46_KWT_01
Test Objective	Verify that the IUT provides a normal forwarding function.
Reference	IETF RFC 7915 [1], clause 4.6
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/1_6_1
Initial Conditions	

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
the IUT is configured with a specific route by which an IPv4 address is reachable within IPv4 pool
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv4 packet with the total length less than 1280 bytes containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address, an IPv4 address is reachable by a specific route
              Data
        from the IPv4 node
    }
    then {
         the IUT forwards that IPv4 packet without translating
         to the IPv4 node
       }
   }
```

5.2.3 Translating from IPv6 to IPv4

5.2.3.1 Translating IPv6 Headers to IPv4 Headers

TP Id	TP_SIIT_T64_IPHDR_01
Test Objective	Verify that the IUT successfully translates IPv6 header into IPv4 header if there is no IPv6 Fragment Header.
Reference	IETF RFC 7915 [1], clause 5.1
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_1_1
Initial Conditions	

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
             Data
        from the IPv6 node
```

```
    }
    then {
         IUT sends an IPv4 packet containing
              IPv4 basic header containing
                  Version, with the value of 4
                  Internet Header Length (IHL), with the value of 5
                  Type of Service (TOS), copied from Traffic Class
                  Total Length, with the value which is payload length value from the IPv6 header,
                                plus the size of the IPv4 header
                  Identification (Fragment ID),
                  Flags, setting the More Fragments flag to 0, and the Don't Fragment (DF) flag
                          is set to 0
                  Fragment Offset, set to all zeros
                  Time to Live (TTL), with the value of decrementing the Hop Limit value
                  Protocol, copied from Next Header
                  Header Checksum, calculating the checksum once the IPv4 header has been created
                  Source Address, mapped to an IPv4 address according the algorithm
                  Destination Address, mapped to an IPv4 address according the algorithm
              Data
         to the IPv4 node
        }
   }
```

TP Id	TP_SIIT_T64_IPHDR_02
Test Objective	Verify that the IUT successfully translates IPv6 header into IPv4 header if there is no IPv6 Fragment Header and the IUT is configured to set the IPv4 TOS Octet to a specific value (5).
Reference	IETF RFC 7915 [1], clause 5.1
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_1_1_2
Initial Conditions	

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class, with the value of 0
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
             Data
        from the IPv6 node
    }
    then {
         IUT sends an IPv4 packet containing
              IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS), with the value of 5
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL),
                  Header Checksum,
                  Destination Address,
              Data
         to the IPv4 node
        }
   }
```

TP Id	TP_SIIT_T64_IPHDR_03
Test Objective	Verify that the IUT successfully translates IPv6 header into IPv4 header if there is no IPv6 Fragment Header and the translated IPv4 packet is greater than 1 260 bytes.
Reference	IETF RFC 7915 [1], clause 5.1
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_1_1_5

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1300 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length equal to 1300 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class, with the value of 0
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
             Data
        from the IPv6 node
    }
    then {
         IUT sends an IPv4 packet containing
              IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags, with Don't Fragment (DF) flag set to 1
                  Fragment Offset,
                  Time to Live (TTL),
                  Header Checksum,
                  Destination Address,
              Data
        to the IPv4 node
       }
   }
```

TP Id	TP_SIIT_T64_IPHDR_04
Test Objective	Verify that the IUT successfully translates IPv6 header into IPv4 header if there is no IPv6 Fragment Header and the Hop Limit value is 0.
Reference	IETF RFC 7915 [1], clause 5.1
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_1_1_7

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value equal to 1
                Source Address,
                Destination Address,
             Data
        from the IPv6 node
    }
    then {
```

```
        IUT sends the ICMpv4 "TTL Exceeded"
        to the IPv4 node
                  or ICMPv6 "Hop Limit Exceeded" error
        to the IPv6 node
      }
  }
```

TP Id	TP_SIIT_T64_IPHDR_05
Test Objective	Verify that the IUT successfully translates IPv6 header into IPv4 header if there is no IPv6 Fragment Header and ignores the IPv6 header HOPOPT (0).
Reference	IETF RFC 7915 [1], clause 5.1
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_1_1_8

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
       }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header, with the value of 0
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
             Hop-by-Hop Options Header containing
                Next Header,
                Hdr Ext Len,
                Options,
             Data
        from the IPv6 node
    }
    then {
         IUT sends an IPv4 packet containing
              IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL),
                  Header Checksum,
                  Destination Address,
               Data
         to the IPv4 node
        }
  }
```

TP Id	TP_SIIT_T64_IPHDR_06
Test Objective	Verify that the IUT successfully translates IPv6 header into IPv4 header if there is no IPv6 Fragment Header and ignores the IPv6 header IPv6-Route (43).
Reference	IETF RFC 7915 [1], clause 5.1
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_1_1_8

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
       }
```

Expected Behaviour

```
ensure that {
    when {
```

```
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header, with the value of 43
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
            Routing Header containing
                Next Header,
                Hdr Ext Len,
                Routing Type,
                Segment Left,
                Type-specific data,
             Data
        from the IPv6 node
    }
    then {
         IUT sends an IPv4 packet containing
              IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL),
                  Header Checksum,
                  Destination Address,
               Data
        to the IPv4 node
       }
   }
```

TP Id	TP_SIIT_T64_IPHDR_07
Test Objective	Verify that the IUT successfully translates IPv6 header into IPv4 header if there is no IPv6 Fragment Header and ignores the IPv6 header IPv6-Opts (60).
Reference	IETF RFC 7915 [1], clause 5.1
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_1_1_8

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header, with the value of 60
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
            Routing Header containing
                Next Header,
                Hdr Ext Len,
                Routing Type,
                Segment Left, with the value of 0
                Type-specific data,
             Data
        from the IPv6 node
    }
    then {
         IUT sends an IPv4 packet containing
              IPv4 basic header containing
                  Version,
```

```
                Internet Header Length (IHL),
                Type of Service (TOS),
                Total Length,
                Identification (Fragment ID),
                Flags,
                Fragment Offset,
                Time to Live (TTL),
                Header Checksum,
                Destination Address,
            Data
       to the IPv4 node
      }
  }
```

TP Id	TP_SIIT_T64_IPHDR_08
Test Objective	Verify that the IUT successfully translates IPv6 header into IPv4 header if there is no IPv6 Fragment Header and ignores any of an IPv6 Hop-by-Hop Options header, Destination Options header, or Routing header with the Segments Left field equal to zero.
Reference	IETF RFC 7915 [1], clause 5.1
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_1_2

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
       }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header, with the value of 60
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
            Routing Header containing
                Next Header,
                Hdr Ext Len,
                Routing Type,
                Segment Left, with the value of 0
                Type-specific data,
             Data
        from the IPv6 node
    }
    then {
         IUT sends an IPv4 packet containing
              IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL),
                  Header Checksum,
                  Destination Address,
              Data
         to the IPv4 node
        }
  }
```

TP Id	TP_SIIT_T64_IPHDR_09
Test Objective	Verify that the IUT successfully translates IPv6 header into IPv4 header if there is no IPv6 Fragment Header and a Routing header with a non-zero Segments Left field (1).
Reference	IETF RFC 7915 [1], clause 5.1
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_1_3

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header, with the value of 60
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
            Routing Header containing
                Next Header,
                Hdr Ext Len,
                Routing Type,
                Segment Left, with the value of 1
                Type-specific data,
             Data
        from the IPv6 node
    }
    then {
         IUT sends an IPv6 packet containing
              IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header, with the value of 60
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
              ICMP header containing
                Type, with the value of 4
                Code, with the value of 0
                Checksum,
                Pointer, with the value of 43
                Data, as much of invoking packet as possible without the ICMPv6 packet exceeding
                      the minimum IPv6 MTU
         to the IPv6 node
        }
   }
```

TP Id	TP_SIIT_T64_IPHDR_10
Test Objective	Verify that the IUT successfully translates IPv6 header into IPv4 header if there is an IPv6 Fragment Header.
Reference	IETF RFC 7915 [1], clause 5.1.1
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_1_4

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
```

```
               Flow Label,
               Payload Length,
               Next Header,
               Hop Limit, with the value greater than or equal to 2
               Source Address,
               Destination Address,
            Fragment Header containing
               Next Header
               Reserved
               Fragment Offset
               Res
               M flag
               Identification
            Data
        from the IPv6 node
    }
    then {
         IUT sends an IPv4 packet containing
              IPv4 basic header containing
                  Version, with the value of 4
                  Internet Header Length (IHL), with the value of 5
                  Type of Service (TOS), copied from Traffic Class
                  Total Length, with the value set to payload length value from the IPv6 header,
                                minus the length of the extension headers up to the Fragmentation
                                Header, minus 8 for the Fragment Header, plus the size of the
                                IPv4 header
                  Identification (Fragment ID), copying the low-order 16 bits in the
                                                Identification field in the Fragment Header
                  Flags, The IPv4 More Fragments (MF) flag is copied from the M flag in the IPv6
                          Fragment Header, and the Don't Fragment (DF) flag is set to 0
                  Fragment Offset, copied from the Fragment Offset field of the IPv6 Fragment
                                   Header
                  Time to Live (TTL), with the value of decrementing the Hop Limit value
                  Protocol, copied from Next Header
                  Header Checksum, calculating the checksum once the IPv4 header has been created
                  Source Address, mapped to an IPv4 address according the algorithm
                  Destination Address, mapped to an IPv4 address according the algorithm
              Data
         to the IPv4 node
        }
   }
```

TP Id	TP_SIIT_T64_IPHDR_11
Test Objective	Verify that the IUT successfully translates IPv6 header into IPv4 header if there is an IPv6 Fragment Header and Next Header field of the Fragment Header is an extension header (except ESP, but including the Authentication Header (AH)).
Reference	IETF RFC 7915 [1], clause 5.1.1
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_1_4_1, PICS_A2/2_1_4_5
Initial Conditions	

with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
}	

Expected Behaviour	

ensure that {
when {
IUT receives an IPv6 packet with the total length less than 1280 bytes containing
IPv6 basic header containing
Version,
Traffic Class,
Flow Label,
Payload Length,
Next Header,
Hop Limit, with the value greater than or equal to 2
Source Address,
Destination Address,
Fragment Header containing
Next Header, with the value of 51
Reserved,
Fragment Offset,
Res,
M flag,
Identification,	

```
            Data
        from the IPv6 node
    }
    then {
         IUT drops that packet
       }
   }
```

TP Id	TP_SIIT_T64_IPHDR_12
Test Objective	Verify that the IUT successfully translates IPv6 header into IPv4 header if there is an IPv6 Fragment Header and the size of IPv4 packet is larger than the MTU of the next-hop interface.
Reference	IETF RFC 7915 [1], clause 5.1.1
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_1_4_7
Initial Conditions	

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
the MTU of the next-hop interface is 576 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with length less than or equal to 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
             Fragment Header containing
                Next Header,
                Reserved,
                Fragment Offset,
                Res,
                M flag,
                Identification,
             Data
        from the IPv6 node
    }
    then {
         IUT translates IPv6 packet
              and fragments IPv4 packet
              and sends packets
        to the IPv4 node
       }
   }
```

5.2.3.2 Translating ICMPv6 Headers into ICMPv4 Headers

TP Id	TP_SIIT_T64_ICMPH_01
Test Objective	Verify that the IUT successfully translates Echo Request messages (128), adjusts the type value to 8 and recalculates the ICMP checksum both to take the type change into account and to exclude the ICMPv6 pseudo-header.
Reference	IETF RFC 7915 [1], clause 5.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_2_1_1
Initial Conditions	

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
```

```
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
            ICMP header containing
                Type, with the value of 128
                Code,
                Checksum,
                Identifier
                Sequence Number,
             Data
        from the IPv6 node
    }
    then {
         IUT sends an IPv4 packet containing
              IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL),
                  Header Checksum,
                  Destination Address,
              ICMP header containing
                  Type, with the value of 8
                  Code,
                  Checksum, adjusting the value both take the type change into account and to
                            exclude the ICMPv6 pseudo-header
                  Identifier,
                  Sequence number,
              Data
         to the IPv4 node
        }
   }
```

TP Id	TP_SIIT_T64_ICMPH_02
Test Objective	Verify that the IUT successfully translates Echo Request messages (129), adjusts the type value to 0 and recalculates the ICMP checksum both to take the type change into account and to exclude the ICMPv6 pseudo-header.
Reference	IETF RFC 7915 [1], clause 5.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_2_1_1

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
            ICMP header containing
                Type, with the value of 129
                Code,
                Checksum,
                Identifier
                Sequence Number,
             Data
        from the IPv6 node
    }
```

```
    then {
         IUT sends an IPv4 packet containing
              IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL),
                  Header Checksum,
                  Destination Address,
              ICMP header containing
                  Type, with the value of 0
                  Code,
                  Checksum, adjusting the value both take the type change into account and to
                            exclude the ICMPv6 pseudo-header
                  Identifier,
                  Sequence number,
              Data
         to the IPv4 node
        }
   }
```

TP Id	TP_SIIT_T64_ICMPH_03
Test Objective	Verify that the IUT successfully drops the MLD Multicast Listener Query messages (130).
Reference	IETF RFC 7915 [1], clause 5.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_2_1_2
Initial Conditions	

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
            ICMP header containing
                Type, with the value of 130
                Code,
                Checksum,
                Maximum Response Delay,
                Reserved,
                Multicast Address
        from the IPv6 node
    }
    then {
         IUT drops that packet
        }
   }
```

TP Id	TP_SIIT_T64_ICMPH_04
Test Objective	Verify that the IUT successfully drops the MLD Multicast Listener Report messages (131).
Reference	IETF RFC 7915 [1], clause 5.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_2_1_2

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
            ICMP header containing
                Type, with the value of 131
                Code,
                Checksum,
                Maximum Response Delay,
                Reserved,
                Multicast Address
        from the IPv6 node
    }
    then {
         IUT drops that packet
       }
   }
```

TP Id	TP_SIIT_T64_ICMPH_04
Test Objective	Verify that the IUT successfully drops the MLD Multicast Listener Done messages (132).
Reference	IETF RFC 7915 [1], clause 5.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_2_1_2

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
            ICMP header containing
                Type, with the value of 132
                Code,
                Checksum,
                Maximum Response Delay,
                Reserved,
                Multicast Address
        from the IPv6 node
    }
    then {
         IUT drops that packet
       }
   }
```

TP Id	TP_SIIT_T64_ICMPH_05
Test Objective	Verify that the IUT successfully drops the Router Solicitation messages (133).
Reference	IETF RFC 7915 [1], clause 5.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_2_1_3

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
            ICMP header containing
                Type, with the value of 133
                Code,
                Checksum,
                Reserved,
                options,
        from the IPv6 node
    }
    then {
         IUT drops that packet
       }
   }
```

TP Id	TP_SIIT_T64_ICMPH_06
Test Objective	Verify that the IUT successfully drops the Router Advertisement messages (134).
Reference	IETF RFC 7915 [1], clause 5.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_2_1_3

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
            ICMP header containing
                Type, with the value of 134
                Code,
                Checksum,
                Cur Hop Limit,
                M,
                O,
                Reserved,
                Router Lifetime,
                Reachable Time,
                Retrans Timer,
                Options,
        from the IPv6 node
    }
    then {
```

```
        IUT drops that packet
      }
  }
```

TP Id	TP_SIIT_T64_ICMPH_07
Test Objective	Verify that the IUT successfully drops the Neighbour Solicitation messages (135).
Reference	IETF RFC 7915 [1], clause 5.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_2_1_3

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
       }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
            ICMP header containing
                Type, with the value of 135
                Code,
                Checksum,
                Reserved,
                Target Address,
                Options,
        from the IPv6 node
    }
    then {
         IUT drops that packet
       }
   }
```

TP Id	TP_SIIT_T64_ICMPH_08
Test Objective	Verify that the IUT successfully drops the Neighbour Advertisement messages (136).
Reference	IETF RFC 7915 [1], clause 5.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_2_1_3

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
       }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
            ICMP header containing
                Type, with the value of 136
                Code,
                Checksum,
                R,
                S,
                O,
                Reserved,
                Target Address,
```

```
                Options,
        from the IPv6 node
    }
    then {
         IUT drops that packet
       }
   }
```

TP Id	TP_SIIT_T64_ICMPH_09
Test Objective	Verify that the IUT successfully drops the Neighbour Advertisement messages (137).
Reference	IETF RFC 7915 [1], clause 5.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_2_1_3
Initial Conditions	

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
            ICMP header containing
                Type, with the value of 137
                Code,
                Checksum,
                Reserved,
                Target Address,
                Destination Address,
                Options,
        from the IPv6 node
    }
    then {
         IUT drops that packet
       }
   }
```

TP Id	TP_SIIT_T64_ICMPH_10
Test Objective	Verify that the IUT successfully drops the Unknown informational messages (254).
Reference	IETF RFC 7915 [1], clause 5.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_2_1_4
Initial Conditions	

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
            ICMP header containing
                Type, with the value of 254
                Code,
                Checksum,
```

```
                Reserved,
                Target Address,
                Destination Address,
                Options,
        from the IPv6 node
    }
    then {
         IUT drops that packet
       }
   }
```

TP Id	TP_SIIT_T64_ICMPH_11
Test Objective	Verify that the IUT successfully adjusts the type values of Destination Unreachable messages (type 1) to 3, the code 0 (No route to destination) to 1 (Host unreachable) and the ICMP checksum both to take the type/code change into account and to exclude the ICMPv6 pseudo-header.
Reference	IETF RFC 7915 [1], clause 5.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_2_2_1_1

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
            ICMP header containing
                Type, with the value of 1
                Code, with the value of 0
                Checksum,
                Reserved,
                Unused,
            Data, as much of invoking packet as possible without the ICMPv6 packet exceeding
                        the minimum IPv6 MTU
        from the IPv6 node
    }
    then {
         IUT sends an IPv4 packet containing
              IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL),
                  Header Checksum,
                  Destination Address,
              ICMP header containing
                  Type, with the value of 3
                  Code, with the value of 1
                  Checksum, adjusting the value both take the type change into account and to
                            exclude the ICMPv6 pseudo-header
                  Unused,
             Data, Internet Header plus 64 bits of Original Data Datagram
        to the IPv4 node
       }
   }
```

TP Id	TP_SIIT_T64_ICMPH_12
Test Objective	Verify that the IUT successfully adjusts the type values of Destination Unreachable messages (type 1) to 3, the code 1 (Communication with destination administratively prohibited) to 10 (Communication with destination host administratively prohibited) and the ICMP checksum both to take the type/code change into account and to exclude the ICMPv6 pseudo-header.
Reference	IETF RFC 7915 [1], clause 5.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_2_2_1_2

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
            ICMP header containing
                Type, with the value of 1
                Code, with the value of 1
                Checksum,
                Reserved,
                Unused,
            Data, as much of invoking packet as possible without the ICMPv6 packet exceeding
                        the minimum IPv6 MTU
        from the IPv6 node
    }
    then {
         IUT sends an IPv4 packet containing
              IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL),
                  Header Checksum,
                  Destination Address,
              ICMP header containing
                  Type, with the value of 3
                  Code, with the value of 10
                  Checksum, adjusting the value both take the type change into account and to
                            exclude the ICMPv6 pseudo-header
                  Unused,
              Data, Internet Header plus 64 bits of Original Data Datagram
        to the IPv4 node
       }
   }
```

TP Id	TP_SIIT_T64_ICMPH_13
Test Objective	Verify that the IUT successfully adjusts the type values of Destination Unreachable messages (type 1) to 3, the code 2 (Beyond scope of source address) to 1 (Host unreachable) and the ICMP checksum both to take the type/code change into account and to exclude the ICMPv6 pseudo-header.
Reference	IETF RFC 7915 [1], clause 5.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_2_2_1_3

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
            ICMP header containing
                Type, with the value of 1
                Code, with the value of 2
                Checksum,
                Reserved,
                Unused,
            Data, as much of invoking packet as possible without the ICMPv6 packet exceeding
                        the minimum IPv6 MTU
        from the IPv6 node
    }
    then {
         IUT sends an IPv4 packet containing
              IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL),
                  Header Checksum,
                  Destination Address,
              ICMP header containing
                  Type, with the value of 3
                  Code, with the value of 1
                  Checksum, adjusting the value both take the type change into account and to
                            exclude the ICMPv6 pseudo-header
                  Unused,
             Data, Internet Header plus 64 bits of Original Data Datagram
        to the IPv4 node
       }
   }
```

TP Id	TP_SIIT_T64_ICMPH_14
Test Objective	Verify that the IUT successfully adjusts the type values of Destination Unreachable messages (type 1) to 3, the code 3 (Address unreachable) to 1 (Host unreachable) and the ICMP checksum both to take the type/code change into account and to exclude the ICMPv6 pseudo-header.
Reference	IETF RFC 7915 [1], clause 5.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_2_2_1_4

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
            ICMP header containing
                Type, with the value of 1
                Code, with the value of 3
```

```
                Checksum,
                Reserved,
                Unused,
            Data, as much of invoking packet as possible without the ICMPv6 packet exceeding
                        the minimum IPv6 MTU
        from the IPv6 node
    }
    then {
         IUT sends an IPv4 packet containing
              IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL),
                  Header Checksum,
                  Destination Address,
              ICMP header containing
                  Type, with the value of 3
                  Code, with the value of 1
                  Checksum, adjusting the value both take the type change into account and to
                            exclude the ICMPv6 pseudo-header
                  Unused,
              Data, Internet Header plus 64 bits of Original Data Datagram
         to the IPv4 node
        }
   }
```

TP Id	TP_SIIT_T64_ICMPH_15
Test Objective	Verify that the IUT successfully adjusts the type values of Destination Unreachable messages (type 1) to 3, the code 4 (Port unreachable) to 3 (Port unreachable) and the ICMP checksum both to take the type/code change into account and to exclude the ICMPv6 pseudo-header.
Reference	IETF RFC 7915 [1], clause 5.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_2_2_1_4

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
            ICMP header containing
                Type, with the value of 1
                Code, with the value of 3
                Checksum,
                Reserved,
                Unused,
            Data, as much of invoking packet as possible without the ICMPv6 packet exceeding
                        the minimum IPv6 MTU
        from the IPv6 node
    }
    then {
         IUT sends an IPv4 packet containing
              IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
```

```
                Fragment Offset,
                Time to Live (TTL),
                Header Checksum,
                Destination Address,
            ICMP header containing
                Type, with the value of 3
                Code, with the value of 3
                Checksum, adjusting the value both take the type change into account and to
                          exclude the ICMPv6 pseudo-header
                Unused,
            Data, Internet Header plus 64 bits of Original Data Datagram
      to the IPv4 node
     }
  }
```

TP Id	TP_SIIT_T64_ICMPH_16
Test Objective	Verify that the IUT successfully drops Destination Unreachable messages with unknown code values (10).
Reference	IETF RFC 7915 [1], clause 5.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_2_2_1_4
Initial Conditions	

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
       }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
            ICMP header containing
                Type, with the value of 1
                Code, with the value of 10
                Checksum,
                Reserved,
                Unused,
            Data, as much of invoking packet as possible without the ICMPv6 packet exceeding
                      the minimum IPv6 MTU
        from the IPv6 node
    }
    then {
         IUT drops that packet
       }
  }
```

TP Id	TP_SIIT_T64_ICMPH_17
Test Objective	Verify that the IUT successfully translates Packet Too Big (Type 2) messages to an ICMPv4 Destination Unreachable (Type 3) with Code 4, and adjust the ICMPv4 checksum both to take the type change into account and to exclude the ICMPv6 pseudo-header.
Reference	IETF RFC 7915 [1], clause 5.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_2_2_2_1
Initial Conditions	

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
       }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
```

```
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
            ICMP header containing
                Type, with the value of 2
                Code, with the value of 0
                Checksum,
                MTU,
                Data, copying as much of invoking packet as possible without the ICMPv6 packet
                      exceeding the minimum IPv6 MTU
        from the IPv6 node
    }
    then {
         IUT sends an IPv4 packet containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL),
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 11
                  Code,
                  Checksum, adjusting the value both take the type change into account and to
                            exclude the ICMPv6 pseudo-header
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
       }
   }
```

TP Id	TP_SIIT_T64_ICMPH_18
Test Objective	Verify that the IUT successfully translates Time Exceeded (Type 3) messages, set the Type to 11, and adjusts the ICMPv4 checksum both to take the type change into account and to exclude the ICMPv6 pseudo-header.
Reference	IETF RFC 7915 [1], clause 5.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_2_2_3
Initial Conditions	

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
            ICMP header containing
                Type, with the value of 3
                Code,
                Checksum,
                Unused,
                Data, copying as much of invoking packet as possible without the ICMPv6 packet
                      exceeding the minimum IPv6 MTU
        from the IPv6 node
```

```
    }
    then {
         IUT sends an IPv4 packet containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL),
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 11
                  Code,
                  Checksum, adjusting the value both take the type change into account and to
                            exclude the ICMPv6 pseudo-header
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
       }
   }
```

TP Id	TP_SIIT_T64_ICMPH_19
Test Objective	Verify that the IUT successfully translates Parameter Problem (Type 4) messages with Code 0 and pointer 0, sets the Type to 12, the Code to 0, and the pointer to 0.
Reference	IETF RFC 7915 [1], clause 5.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_2_2_4_1

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
            ICMP header containing
                Type, with the value of 4
                Code, with the value of 0
                Checksum,
                Pointer, with the value of 0
                Data, copying as much of invoking packet as possible without the ICMPv6 packet
                      exceeding the minimum IPv6 MTU
        from the IPv6 node
    }
    then {
         IUT sends an IPv4 packet containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL),
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
```

```
            ICMP header containing
                 Type, with the value of 12
                 Code, with the value of 0
                 Checksum, adjusting the value both take the type change into account and to
                           exclude the ICMPv6 pseudo-header
                 Pointer, with the value of 0
                 unused,
                 Internet Header + 64 bits of Original Data Datagram
       from the IPv4 node
      }
  }
```

TP Id	TP_SIIT_T64_ICMPH_20
Test Objective	Verify that the IUT successfully translates Parameter Problem (Type 4) messages with Code 0 and pointer 1, sets the Type to 12, the Code to 0, and the pointer to 1.
Reference	IETF RFC 7915 [1], clause 5.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_2_2_4_1

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
       }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
            ICMP header containing
                Type, with the value of 4
                Code, with the value of 0
                Checksum,
                Pointer, with the value of 1
                Data, copying as much of invoking packet as possible without the ICMPv6 packet
                      exceeding the minimum IPv6 MTU
        from the IPv6 node
    }
    then {
         IUT sends an IPv4 packet containing
            IPv4 basic header containing
                 Version,
                 Internet Header Length (IHL),
                 Type of Service (TOS),
                 Total Length,
                 Identification (Fragment ID),
                 Flags,
                 Fragment Offset,
                 Time to Live (TTL),
                 Protocol,
                 Header Checksum,
                 Source Address,
                 Destination Address,
            ICMP header containing
                 Type, with the value of 12
                 Code, with the value of 0
                 Checksum, adjusting the value both take the type change into account and to
                           exclude the ICMPv6 pseudo-header
                 Pointer, with the value of 1
                 unused,
                 Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
       }
  }
```

TP Id	TP_SIIT_T64_ICMPH_21
Test Objective	Verify that the IUT successfully drops Parameter Problem (Type 4) messages with Code 0 and pointer 2.
Reference	IETF RFC 7915 [1], clause 5.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_2_2_4_1

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
            ICMP header containing
                Type, with the value of 4
                Code, with the value of 0
                Checksum,
                Pointer, with the value of 2
                Data, copying as much of invoking packet as possible without the ICMPv6 packet
                      exceeding the minimum IPv6 MTU
        from the IPv6 node
    }
    then {
         IUT drops that packet
       }
   }
```

TP Id	TP_SIIT_T64_ICMPH_22
Test Objective	Verify that the IUT successfully drops Parameter Problem (Type 4) messages with Code 0 and pointer 3.
Reference	IETF RFC 7915 [1], clause 5.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_2_2_4_1

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
            ICMP header containing
                Type, with the value of 4
                Code, with the value of 0
                Checksum,
                Pointer, with the value of 3
                Data, copying as much of invoking packet as possible without the ICMPv6 packet
                      exceeding the minimum IPv6 MTU
        from the IPv6 node
    }
    then {
         IUT drops that packet
```

```
      }
   }
```

TP Id	TP_SIIT_T64_ICMPH_23
Test Objective	Verify that the IUT successfully translates Parameter Problem (Type 4) messages with Code 0 and pointer 4, sets the Type to 12, the Code to 0, and the pointer to 2.
Reference	IETF RFC 7915 [1], clause 5.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_2_2_4_1

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
       }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
            ICMP header containing
                Type, with the value of 4
                Code, with the value of 0
                Checksum,
                Pointer, with the value of 4
                Data, copying as much of invoking packet as possible without the ICMPv6 packet
                      exceeding the minimum IPv6 MTU
        from the IPv6 node
    }
    then {
         IUT sends an IPv4 packet containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL),
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 12
                  Code, with the value of 0
                  Checksum, adjusting the value both take the type change into account and to
                            exclude the ICMPv6 pseudo-header
                  Pointer, with the value of 2
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
       }
  }
```

TP Id	TP_SIIT_T64_ICMPH_24
Test Objective	Verify that the IUT successfully translates Parameter Problem (Type 4) messages with Code 0 and pointer 5, sets the Type to 12, the Code to 0, and the pointer to 2.
Reference	IETF RFC 7915 [1], clause 5.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_2_2_4_1

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
            ICMP header containing
                Type, with the value of 4
                Code, with the value of 0
                Checksum,
                Pointer, with the value of 5
                Data, copying as much of invoking packet as possible without the ICMPv6 packet
                      exceeding the minimum IPv6 MTU
        from the IPv6 node
    }
    then {
         IUT sends an IPv4 packet containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL),
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 12
                  Code, with the value of 0
                  Checksum, adjusting the value both take the type change into account and to
                            exclude the ICMPv6 pseudo-header
                  Pointer, with the value of 2
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
         from the IPv4 node
        }
   }
```

TP Id	TP_SIIT_T64_ICMPH_25
Test Objective	Verify that the IUT successfully translates Parameter Problem (Type 4) messages with Code 0 and pointer 6, sets the Type to 12, the Code to 0, and the pointer to 9.
Reference	IETF RFC 7915 [1], clause 5.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_2_2_4_1

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
```

```
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
            ICMP header containing
                Type, with the value of 4
                Code, with the value of 0
                Checksum,
                Pointer, with the value of 6
                Data, copying as much of invoking packet as possible without the ICMPv6 packet
                      exceeding the minimum IPv6 MTU
        from the IPv6 node
    }
    then {
         IUT sends an IPv4 packet containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL),
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 12
                  Code, with the value of 0
                  Checksum, adjusting the value both take the type change into account and to
                            exclude the ICMPv6 pseudo-header
                  Pointer, with the value of 9
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
       }
   }
```

TP Id	TP_SIIT_T64_ICMPH_26
Test Objective	Verify that the IUT successfully translates Parameter Problem (Type 4) messages with Code 0 and pointer 7, sets the Type to 12, the Code to 0, and the pointer to 8.
Reference	IETF RFC 7915 [1], clause 5.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_2_2_4_1
Initial Conditions	

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
            ICMP header containing
                Type, with the value of 4
                Code, with the value of 0
                Checksum,
                Pointer, with the value of 7
```

```
                Data, copying as much of invoking packet as possible without the ICMPv6 packet
                      exceeding the minimum IPv6 MTU
        from the IPv6 node
    }
    then {
         IUT sends an IPv4 packet containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL),
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 12
                  Code, with the value of 0
                  Checksum, adjusting the value both take the type change into account and to
                            exclude the ICMPv6 pseudo-header
                  Pointer, with the value of 8
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
       }
   }
```

TP Id	TP_SIIT_T64_ICMPH_27
Test Objective	Verify that the IUT successfully translates Parameter Problem (Type 4) messages with Code 0 and pointer 7, sets the Type to 12, the Code to 0, and the pointer to 8.
Reference	IETF RFC 7915 [1], clause 5.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_2_2_4_1

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
            ICMP header containing
                Type, with the value of 4
                Code, with the value of 0
                Checksum,
                Pointer, with the value of 7
                Data, copying as much of invoking packet as possible without the ICMPv6 packet
                      exceeding the minimum IPv6 MTU
        from the IPv6 node
    }
    then {
         IUT sends an IPv4 packet containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL),
```

```
                Protocol,
                Header Checksum,
                Source Address,
                Destination Address,
           ICMP header containing
                Type, with the value of 12
                Code, with the value of 0
                Checksum, adjusting the value both take the type change into account and to
                          exclude the ICMPv6 pseudo-header
                Pointer, with the value of 8
                unused,
                Internet Header + 64 bits of Original Data Datagram
      from the IPv4 node
     }
  }
```

TP Id	TP_SIIT_T64_ICMPH_28
Test Objective	Verify that the IUT successfully translates Parameter Problem (Type 4) messages with Code 0 and pointer ranging from 8 to 23, sets the Type to 12, the Code to 0, and the pointer to 12.
Reference	IETF RFC 7915 [1], clause 5.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_2_2_4_1
Initial Conditions	

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
       }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
            ICMP header containing
                Type, with the value of 4
                Code, with the value of 0
                Checksum,
                Pointer, with the value ranging from 8 to 23
                Data, copying as much of invoking packet as possible without the ICMPv6 packet
                      exceeding the minimum IPv6 MTU
        from the IPv6 node
    }
    then {
         IUT sends an IPv4 packet containing
            IPv4 basic header containing
                 Version,
                 Internet Header Length (IHL),
                 Type of Service (TOS),
                 Total Length,
                 Identification (Fragment ID),
                 Flags,
                 Fragment Offset,
                 Time to Live (TTL),
                 Protocol,
                 Header Checksum,
                 Source Address,
                 Destination Address,
            ICMP header containing
                 Type, with the value of 12
                 Code, with the value of 0
                 Checksum, adjusting the value both take the type change into account and to
                           exclude the ICMPv6 pseudo-header
                 Pointer, with the value of 12
                 unused,
                 Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
       }
  }
```

TP Id	TP_SIIT_T64_ICMPH_29
Test Objective	Verify that the IUT successfully translates Parameter Problem (Type 4) messages with Code 0 and pointer ranging from 24 to 39, sets the Type to 12, the Code to 0, and the pointer to 16.
Reference	IETF RFC 7915 [1], clause 5.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_2_2_4_1

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
            ICMP header containing
                Type, with the value of 4
                Code, with the value of 0
                Checksum,
                Pointer, with the value ranging from 24 to 39
                Data, copying as much of invoking packet as possible without the ICMPv6 packet
                      exceeding the minimum IPv6 MTU
        from the IPv6 node
    }
    then {
         IUT sends an IPv4 packet containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL),
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 12
                  Code, with the value of 0
                  Checksum, adjusting the value both take the type change into account and to
                            exclude the ICMPv6 pseudo-header
                  Pointer, with the value of 16
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
       }
   }
```

TP Id	TP_SIIT_T64_ICMPH_30
Test Objective	Verify that the IUT successfully translates Parameter Problem (Type 4) messages with Code 1, sets the Type to 3, and the Code to 2.
Reference	IETF RFC 7915 [1], clause 5.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_2_2_4_2

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
```

```
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
            ICMP header containing
                Type, with the value of 4
                Code, with the value of 0
                Checksum,
                Pointer, with the value ranging from 8 to 23
                Data, copying as much of invoking packet as possible without the ICMPv6 packet
                      exceeding the minimum IPv6 MTU
        from the IPv6 node
    }
    then {
         IUT sends an IPv4 packet containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL),
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
              ICMP header containing
                  Type, with the value of 12
                  Code, with the value of 0
                  Checksum, adjusting the value both take the type change into account and to
                            exclude the ICMPv6 pseudo-header
                  Pointer, with the value of 12
                  unused,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
       }
   }
```

TP Id	TP_SIIT_T64_ICMPH_31
Test Objective	Verify that the IUT successfully drops Unknown error messages (type 0).
Reference	IETF RFC 7915 [1], clause 5.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_2_2_5
Initial Conditions	

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
            ICMP header containing
                Type, with the value of 0
                Code,
                Checksum,
                Unused,
                Data, copying as much of invoking packet as possible without the ICMPv6 packet
```

```
                    exceeding the minimum IPv6 MTU
        from the IPv6 node
    }
    then {
         IUT drops that packet
       }
   }
```

TP Id	TP_SIIT_T64_ICMPH_32
Test Objective	Verify that the IUT successfully translates ICMPv6 packet contains an ICMPv6 Extension, adjust the ICMPv6 Extension length attribute accordingly.
Reference	IETF RFC 7915 [1], clause 5.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_2_2_6
Initial Conditions	

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
            ICMP header containing
                Type, with the value of 1
                Code, with the value of 0
                Checksum,
                Length,
                Unused,
                Data, copying as much of invoking packet as possible without the ICMPv6 packet
                      exceeding the minimum IPv6 MTU
        from the IPv6 node
    }
    then {
         IUT sends an IPv4 packet containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL),
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 3
                  Code, with the value of 1
                  Checksum, adjusting the value both take the type change into account and to
                            exclude the ICMPv6 pseudo-header
                  Unused,
                  Length, adjusting the value due to changes caused by translation
                  Next-Hop MTU,
                  Internet Header + 64 bits of Original Data Datagram
        from the IPv4 node
       }
   }
```

TP Id	TP_SIIT_T64_ICMPH_34
Test Objective	Verify that the IUT successfully passes ICMPv6 packet with extensions not defined in IETF RFC 4884 [4] as opaque bit strings and any IPv6 address literals contained.
Reference	IETF RFC 7915 [1], clause 5.2
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_2_2_7

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address,
            ICMP header containing
                Type, with the value of 1
                Code, with the value of 0
                Checksum,
                Length,
                Unused,
                Pointer, the field that not defined
                Data, copying as much of invoking packet as possible without the ICMPv6 packet
                      exceeding the minimum IPv6 MTU
                Extension Header containing
                      Version,
                      Reserved,
                      Checksum,
                Extension object containing
                      Length,
                      Class-Num,
                      C-Type,
                      Object payload,
        from the IPv6 node
    }
    then {
         IUT sends an IPv4 packet containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL),
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 3
                  Code, with the value of 1
                  Checksum, adjusting the value both take the type change into account and to
                            exclude the ICMPv6 pseudo-header
                  Unused,
                  Length, adjusting the value due to changes caused by translation
                  Next-Hop MTU,
                  Internet Header + 64 bits of Original Data Datagram
                  Data, copied from the ICMP Extension without translation
        from the IPv4 node
       }
   }
```

5.2.3.3 Translating ICMPv6 Error Messages into ICMPv4

TP Id	TP_SIIT_T64_ICMPE_01
Test Objective	Verify the IUT translates the ICMP error messages containing the packet in error just like a normal IP packet (except that the TTL/Hop Limit value of the inner IPv4/IPv6 packet are not decremented), and update the Total Length field in the outer IPv4 header.
Reference	IETF RFC 7915 [1], clause 5.3
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_3_1

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
       }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an IPv6 packet with the total length less than 1280 bytes containing
             IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length, with value of taking into account the length change in Data
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             ICMP Header containing
                Type, with the value of 1
                Code, with the value of 0
                Checksum,
                Unused,
                Data containing
                     IPv6 basic header containing
                          Version,
                          Traffic Class,
                          Flow Label,
                          Payload Length,
                          Next Header,
                          Hop Limit,
                          Source Address,
                          Destination Address,
                     Data,
        from the IPv6 node
    }
    then {
         the IUT sends an IPv4 packet containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 3
                  Code, with the value of 0
                  Checksum,
                  Pointer,
                  unused,
                  Internet Header + 64 bits of Original Data Datagram containing
                         IPv4 basic header containing
                              Version,
                              Internet Header Length (IHL),
                              Type of Service (TOS),
                              Total Length,
                              Identification (Fragment ID),
                              Flags,
                              Fragment Offset,
                              Time to Live (TTL), no changes
```

```
                              Protocol,
                              Header Checksum,
                              Source Address, mapped to an IPv4 address according the algorithm
                              Destination Address, mapped to an IPv4 address according the
                                                   algorithm
                         Data Datagram
        to the IPv6 node
       }
  }
```

TP Id	TP_SIIT_T64_ICMPE_02
Test Objective	Verify the IUT translates the inner IP header.
Reference	IETF RFC 7915 [1], clause 5.3
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_3_2

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
       }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an IPv6 packet with the total length less than 1280 bytes containing
             IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             ICMP Header containing
                Type, with the value of 1
                Code, with the value of 0
                Checksum,
                Unused,
                Data containing
                     IPv6 basic header containing
                          Version,
                          Traffic Class,
                          Flow Label,
                          Payload Length,
                          Next Header,
                          Hop Limit,
                          Source Address,
                          Destination Address,
                     Data, without embedded IP headers
        from the IPv6 node
    }
    then {
         the IUT sends an IPv4 packet containing
             IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL), with the value greater than or equal to 2
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             ICMP header containing
                  Type, with the value of 3
                  Code, with the value of 0
                  Checksum,
                  Pointer,
                  unused,
                  Internet Header + 64 bits of Original Data Datagram containing
                         IPv4 basic header containing
```

```
                              Version,
                              Internet Header Length (IHL),
                              Type of Service (TOS),
                              Total Length,
                              Identification (Fragment ID),
                              Flags,
                              Fragment Offset,
                              Time to Live (TTL), with the value greater than or equal to 2
                              Protocol,
                              Header Checksum,
                              Source Address,
                              Destination Address,
                         Data Datagram,

       to the IPv4 node
      }
  }
```

TP Id	TP_SIIT_T64_ICMPE_03
Test Objective	Verify the IUT stops the process of translating the outer IP headers at the first embedded header and drops the packet if it contains more embedded headers.
Reference	IETF RFC 7915 [1], clause 5.3
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_3_2

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
       }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
             IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             ICMP Header containing
                Type, with the value of 1
                Code, with the value of 0
                Checksum,
                Unused,
                Data containing
                     IPv6 basic header containing
                          Version,
                          Traffic Class,
                          Flow Label,
                          Payload Length,
                          Next Header,
                          Hop Limit,
                          Source Address,
                          Destination Address,
                     Data, with embedded IP headers
        from the IPv6 node
    }
    then {
         the IUT drops that IPv6 packet
       }
  }
```

5.2.3.4 Generation of ICMPv6 Error Messages

TP Id	TP_SIIT_T64_GE_01
Test Objective	Verify that the IUT sends back an ICMPv6 error message to the original sender of the packet if the IPv6 packet is discarded.
Reference	IETF RFC 7915 [1], clause 5.4
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_4_1, 2_4_2

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
the IUT is configured to the access of IPv6 node is denied
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
             IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
                Source Address,
                Destination Address,
             Data
        from the IPv6 node
    }
    then {
        IUT discards that IPv6 packet and
             sends an IPv6 packet with the total length less than 1280 bytes containing
             IPv6 basic header containing
                  Version,
                  Traffic Class,
                  Flow Label,
                  Payload Length,
                  Next Header,
                  Hop Limit,
                  Source Address,
                  Destination Address, with the IPv6 address of that IPv6 packet
             ICMP header containing
                  Type, with the value of 1
                  Code, with the value of 1
                  Checksum,
                  unused,
                  Data
        to the IPv6 node
       }
   }
```

5.2.3.5 Transport-Layer Header Translation

TP Id	TP_SIIT_T64_TLH_01
Test Objective	Verify that IUT recalculates and updates TCP headers when the address translation algorithm is not checksum neutral.
Reference	IETF RFC 7915 [1], clause 5.5
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_5_1

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
             IPv6 basic header containing
                Version,
                Traffic Class,
```

```
               Flow Label,
               Payload Length,
               Next Header,
               Hop Limit, with the value greater than or equal to 2
               Source Address,
               Destination Address,
            TCP header containing
                Source Port,
                Destination Port,
                Sequence Number,
                Acknowledge Number,
                Data Offset,
                Reserved,
                Control Bits,
                Window,
                Checksum,
                Urgent Point,
                Options,
             Data
        from the IPv6 node
    }
    then {
         the IUT sends an IPv4 packet containing
            IPv4 basic header containing
                  Version,
                  Internet Header Length (IHL),
                  Type of Service (TOS),
                  Total Length,
                  Identification (Fragment ID),
                  Flags,
                  Fragment Offset,
                  Time to Live (TTL),
                  Protocol,
                  Header Checksum,
                  Source Address,
                  Destination Address,
             TCP header containing
                 Source Port,
                 Destination Port,
                 Sequence Number,
                 Acknowledge Number,
                 Data Offset,
                 Reserved,
                 Control Bits,
                 Window,
                 Checksum, with the value taking into account IP addresses change
                 Urgent Point,
                 Options,
              Data
         to the IPv4 node
        }
   }
```

TP Id	TP_SIIT_T64_TLH_02
Test Objective	Verify that IUT recalculates and updates UDP headers when the checksum value is not zero and the address translation algorithm is not checksum neutral.
Reference	IETF RFC 7915 [1], clause 5.5
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_5_1
Initial Conditions	

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
        }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit,
```

```
               Source Address,
               Destination Address,
            UDP header containing
                Source Port,
                Destination Port,
                Sequence Number,
                Checksum, with the value greater than 0
            Data
        from the IPv6 node
    }
    then {
         the IUT sends an IPv4 packet containing
            IPv4 basic header containing
                 Version,
                 Internet Header Length (IHL),
                 Type of Service (TOS),
                 Total Length,
                 Identification (Fragment ID),
                 Flags,
                 Fragment Offset,
                 Time to Live (TTL), with the value greater than or equal to 2
                 Protocol,
                 Header Checksum,
                 Source Address,
                 Destination Address,
            UDP header containing
                Source Port,
                Destination Port,
                Length,
                Checksum, with the value taking into account IP addresses change
             Data
        to the IPv4 node
       }
   }
```

5.2.3.6 Knowing When to Translate

TP Id	TP_SIIT_T64_KWT_01
Test Objective	Verify that the IUT provides a normal forwarding function.
Reference	IETF RFC 7915 [1], clause 5.6
Configuration	CF_XLAT_SIIT
PICS Selection	PICS_A2/2_6_1

Initial Conditions

```
with {
the IUT is configured with the IPv6 prefix and the IPv4 address pool
the value of the minimum IPv6 MTU in the network is 1280 bytes
the IUT is configured with a specific route by which an IPv6 address is reachable within IPv6 pool
       }
```

Expected Behaviour

```
ensure that {
    when {
        IUT receives an IPv6 packet with the total length less than 1280 bytes containing
             IPv6 basic header containing
            IPv6 basic header containing
                Version,
                Traffic Class,
                Flow Label,
                Payload Length,
                Next Header,
                Hop Limit, with the value greater than or equal to 2
                Source Address,
                Destination Address, an IPv6 address is reachable by a specific route
             Data
        from the IPv6 node
    }
    then {
         the IUT forwards that IPv6 packet without translating
         to the IPv6 node
       }
   }
```

History

Version	Date	Status
V1.1.1	February 2026	Publication